

INTRODUCTION À L'APPRENTISSAGE AUTOMATIQUE POUR LES GEOSCIENCES

Jérémy Rohmer, BRGM, j.rohmer@brgm.fr

2 octobre 2026

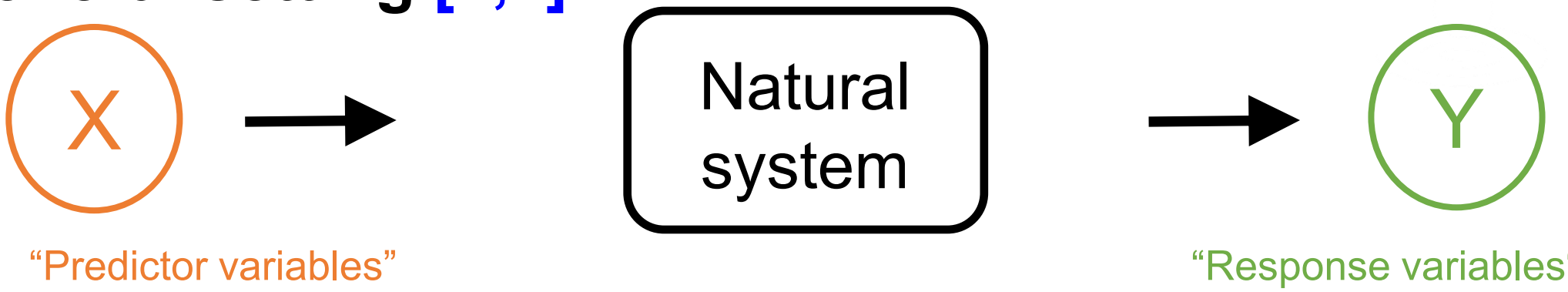


#01

CONTEXTE

Géosciences pour une Terre durable
brgm

General setting [1,2]

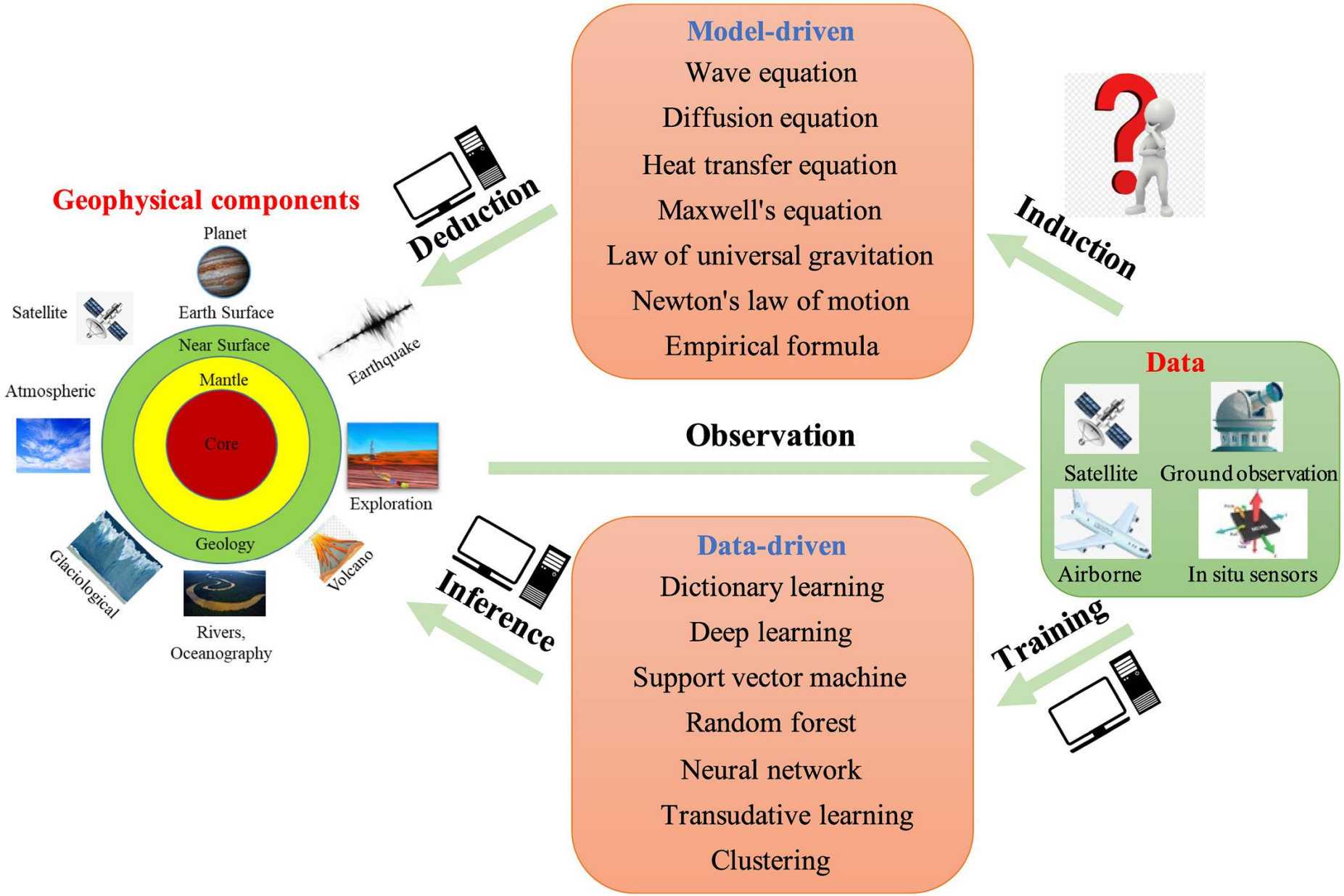


Mathematically $y = f(X)$

- **Science:** Extract information about the law of nature—the function *f.*
- **Prediction:** Predict what the response variables Y are going to be with the predictor variables X revealed to us.
- **Numerical simulators or Machine Learning (ML) tools (denoted g)** try to quantify the relationship under "nature" creating an input output mapping:

$$y = f(X) \approx g(X)$$

[1] Zhao & Hastie (2021); **[2]** Borgonovo (2021)

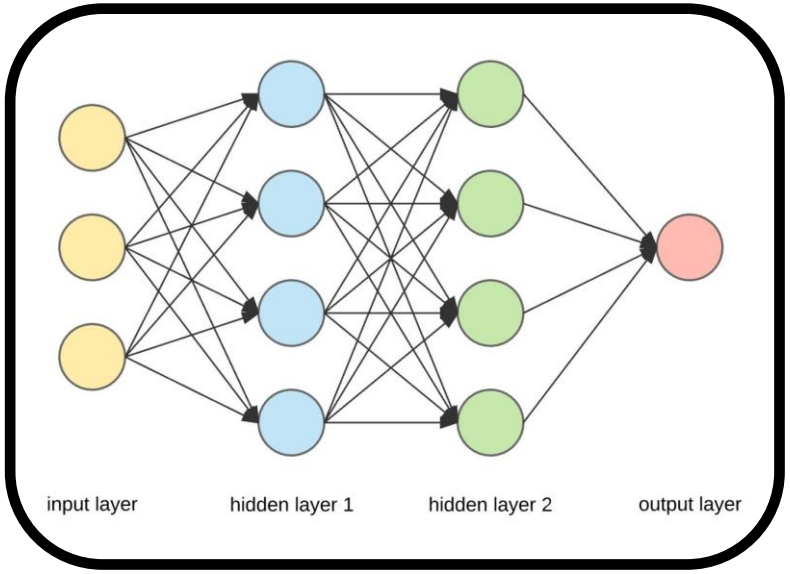


Yu and Ma (2021)

Mineral prospectivity

X

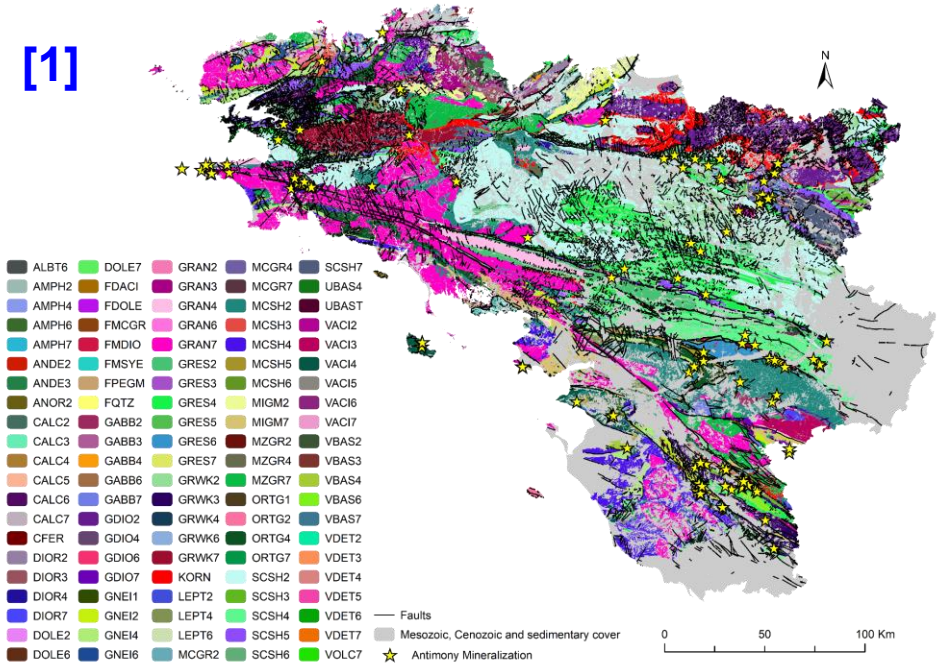
Punctual observations (mineralization) + spatial predictors (geological map, geophysical measurements, etc.)



Y

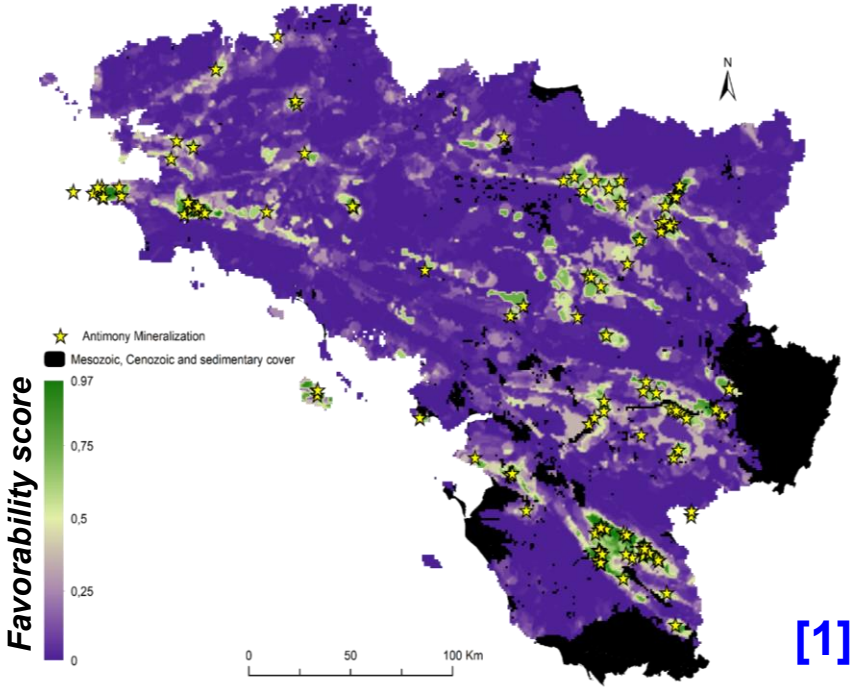
Favorability map (~ probability of mineralization)

[1]



g

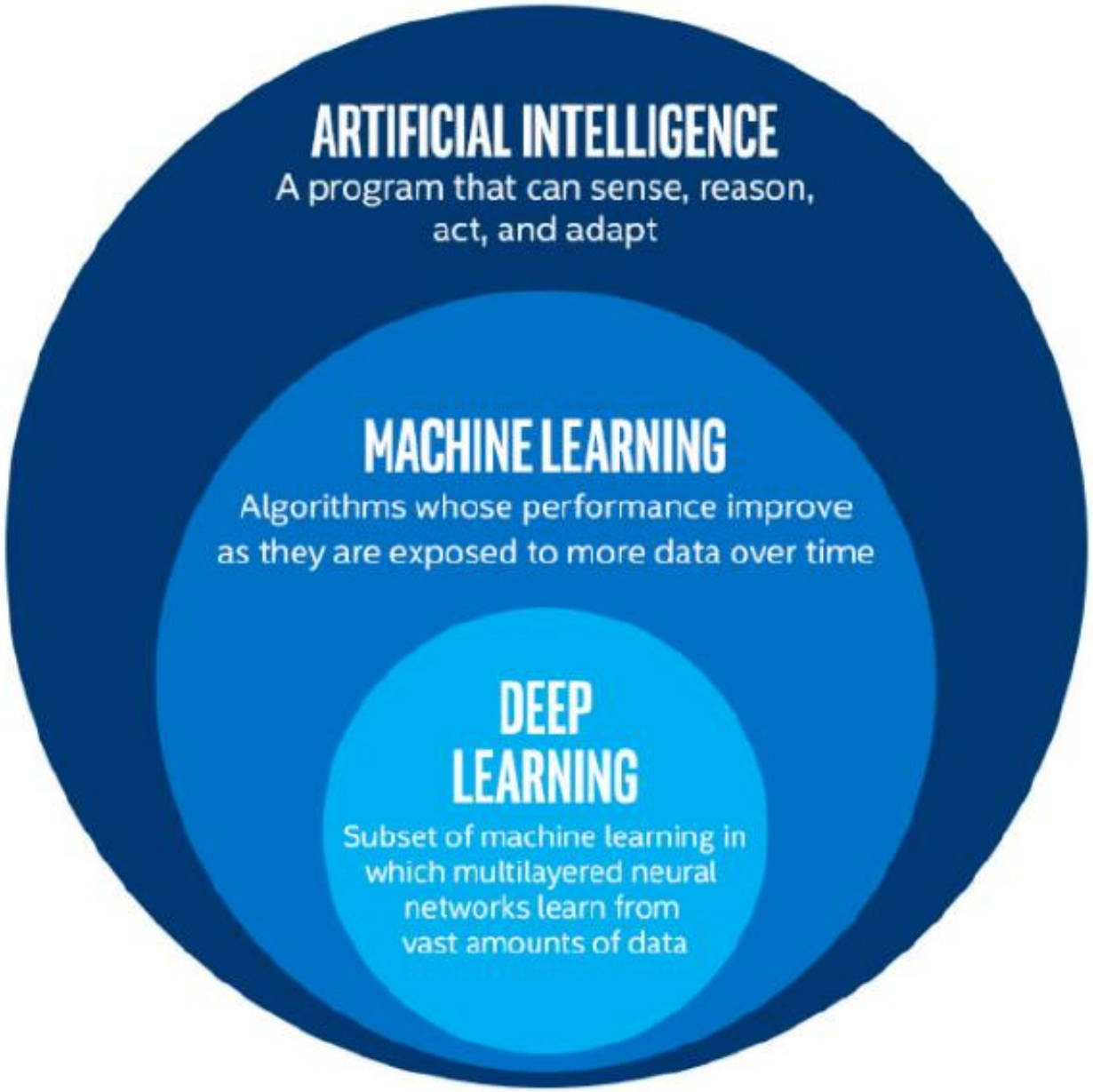
Machine/deep learning method



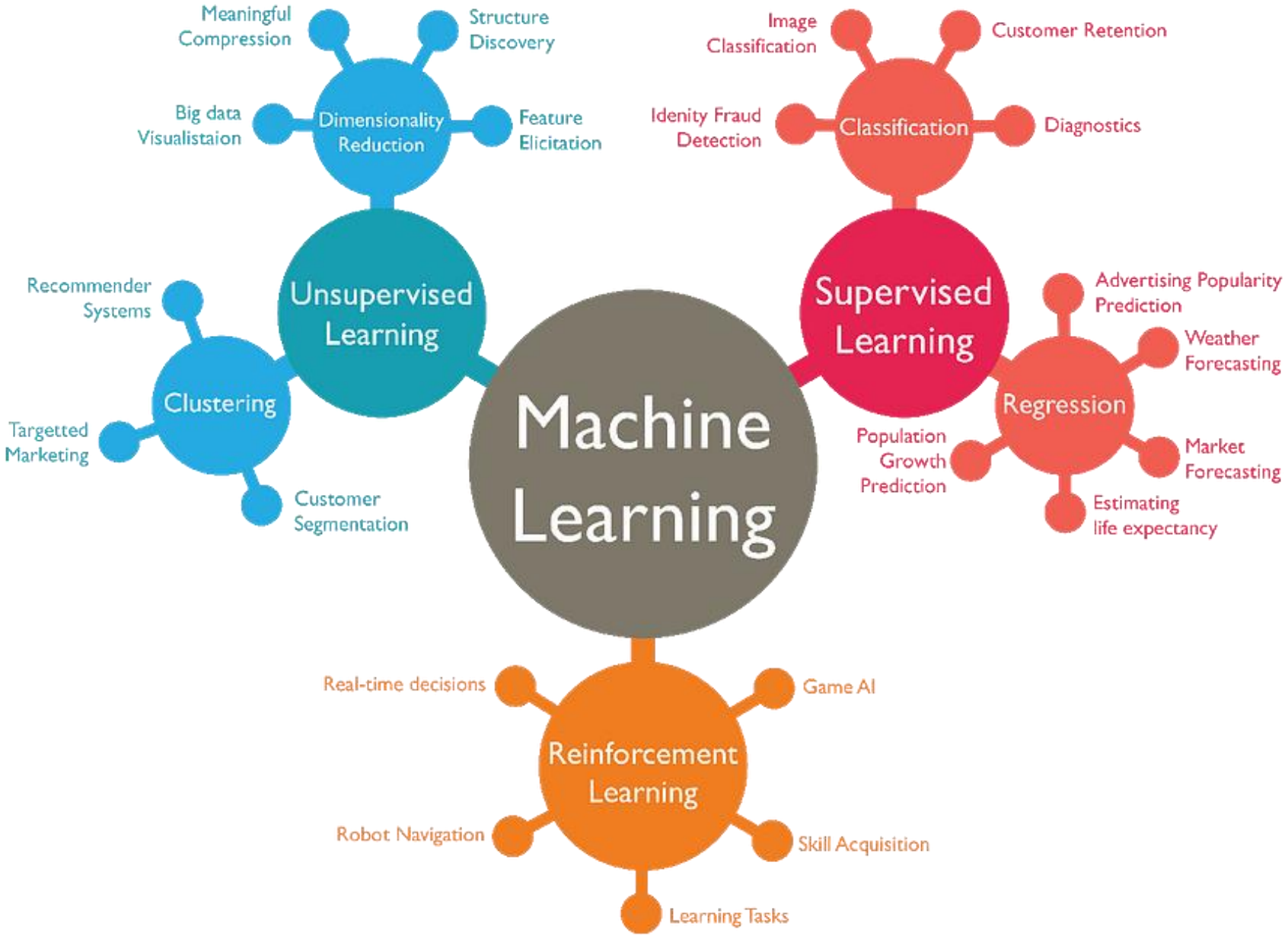
[1]

[1] Vella and co-authors, 2022

Machine learning ?



Machine learning ?

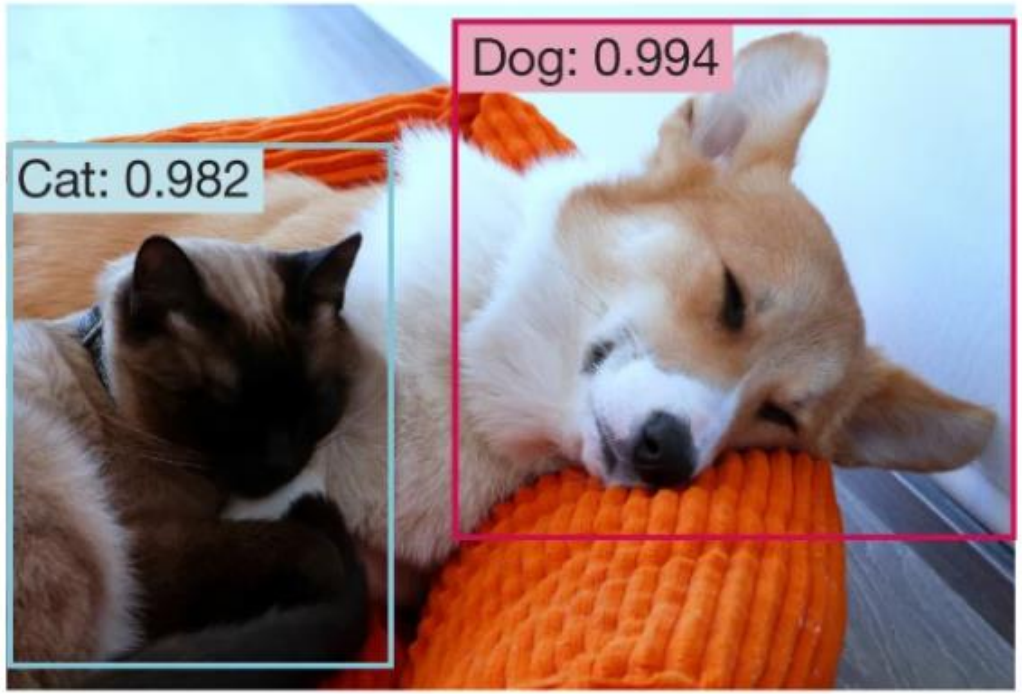


https://medium.com/@victorfanjos/what-is-ai-specifically-what-is-machine-learning-ae155f31351b

Machine learning for Earth sciences

Machine learning tasks

a **Object classification and localization**



Earth science tasks

Pattern classification



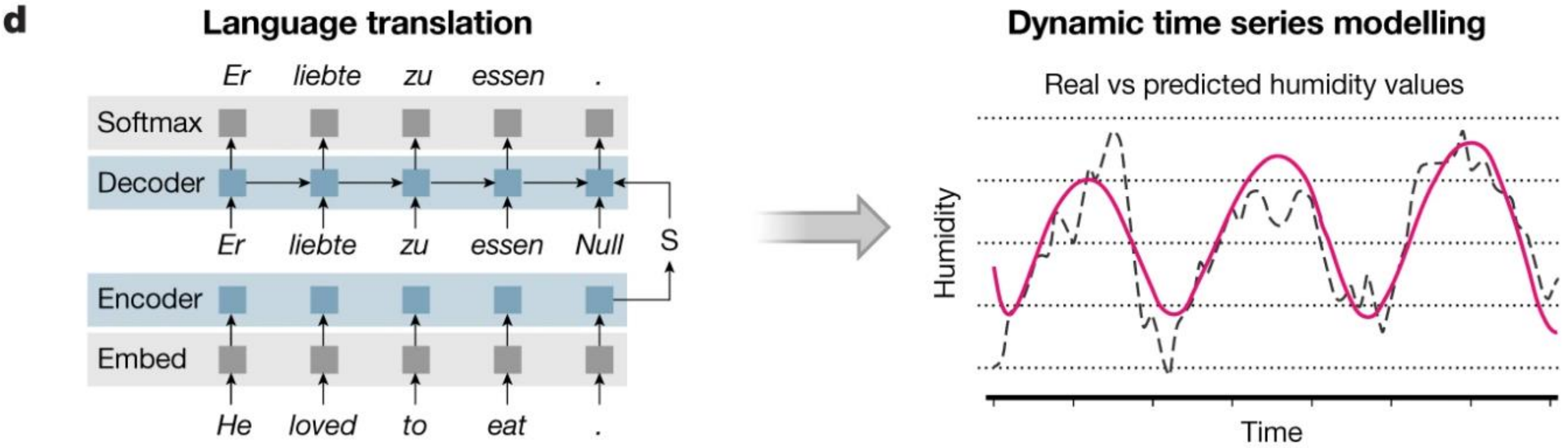
Water classification from MODIS (NASA) satellite images

Reichstein et al. (2019); https://www.earthdata.nasa.gov/learn/trainings/fundamentals-machine-learning-earth-science

Machine learning for Earth sciences

Machine learning tasks

Earth science tasks



Reichstein et al. (2019)

Machine learning for Earth sciences

Machine learning tasks

Earth science tasks

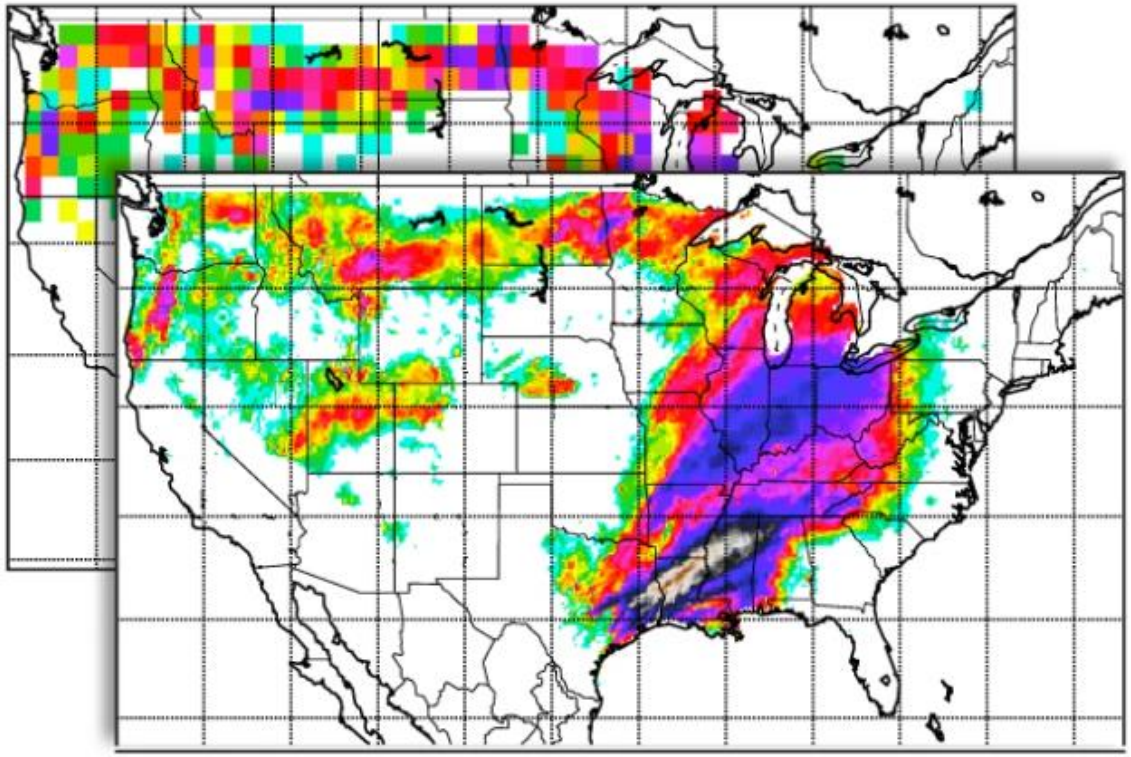
b **Super-resolution and fusion**

8 × 8 input | 32 × 32 samples | Ground truth



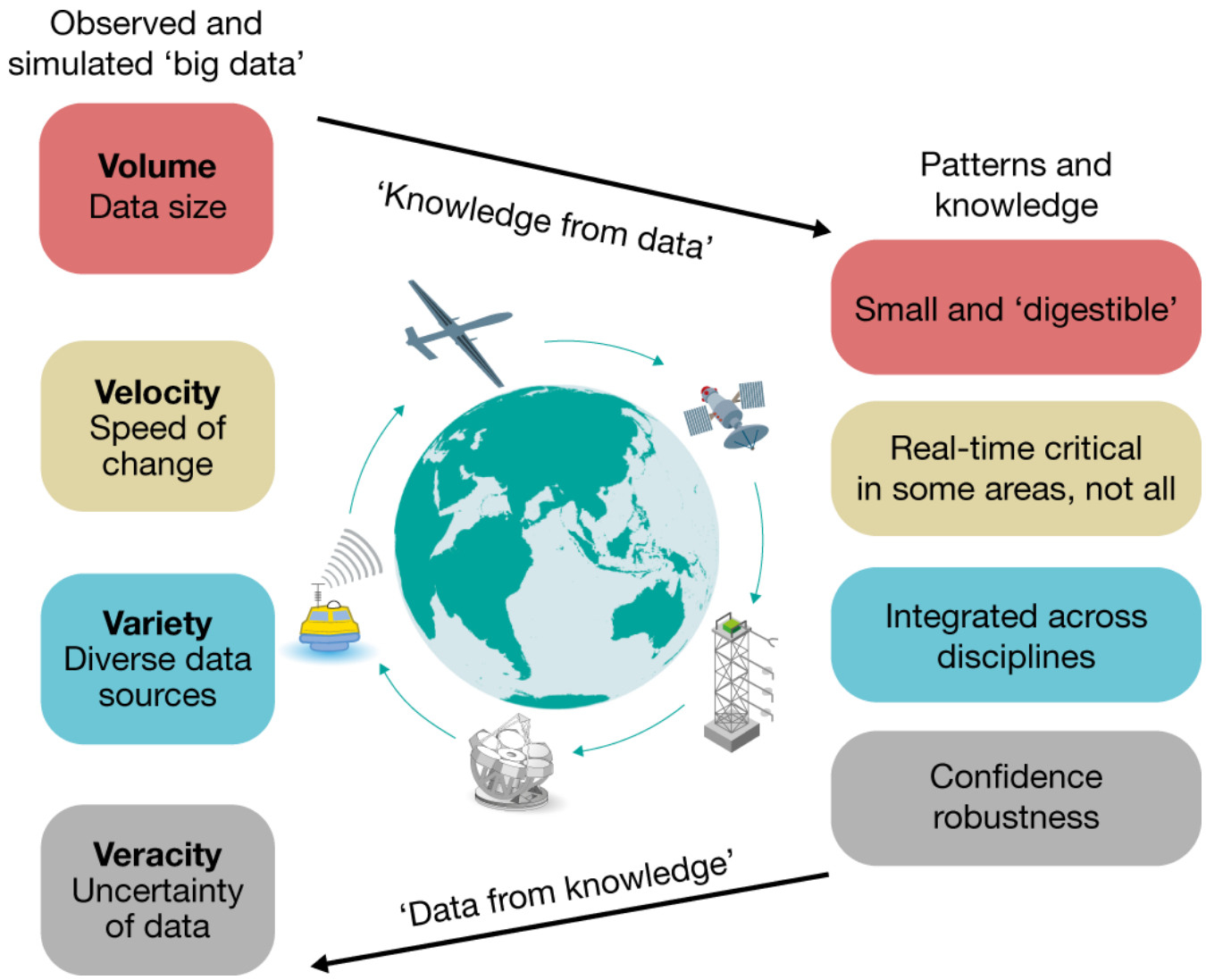
Precipitation

Statistical downscaling and blending



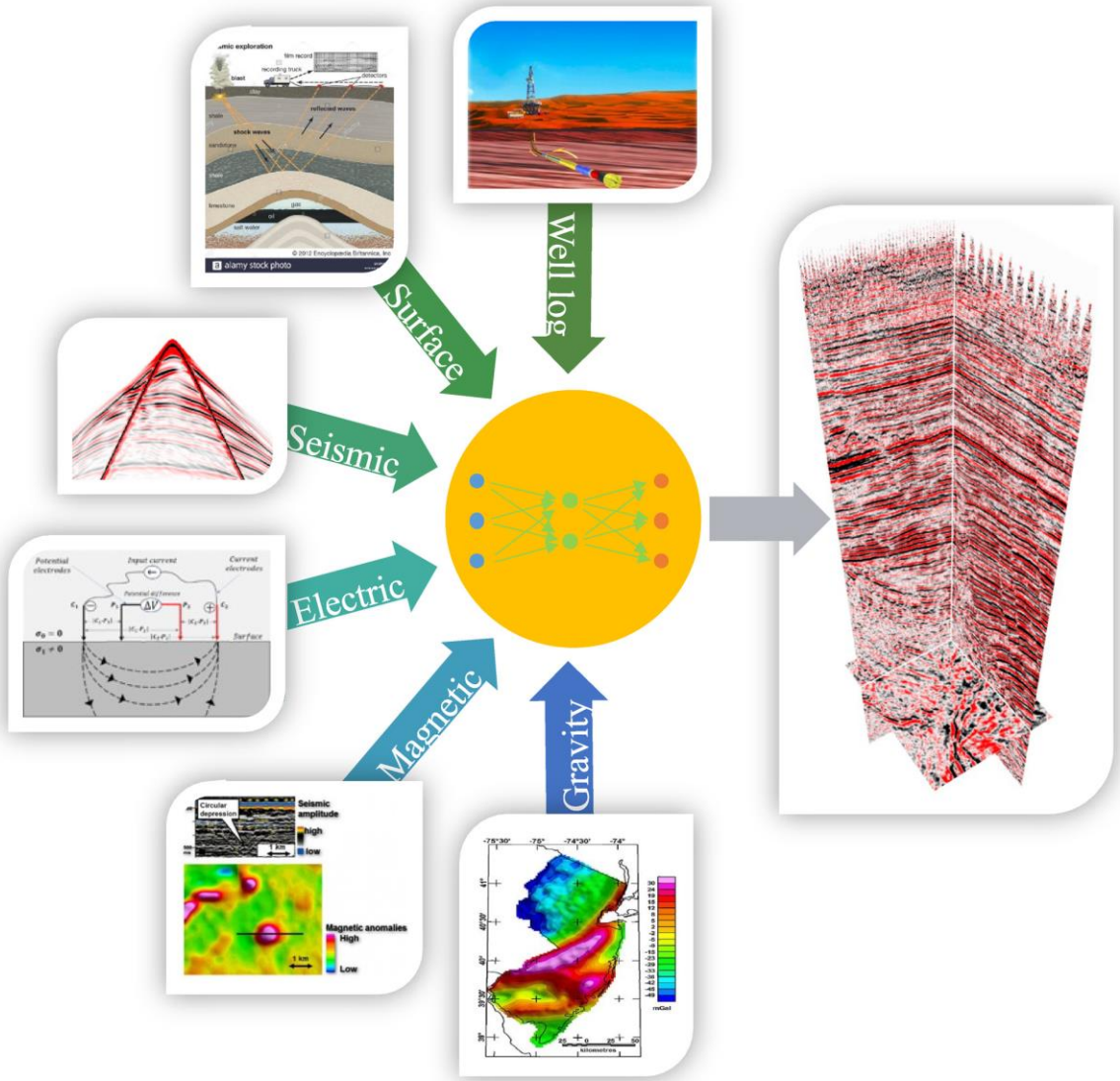
Reichstein et al. (2019)

Challenges in Earth Sciences



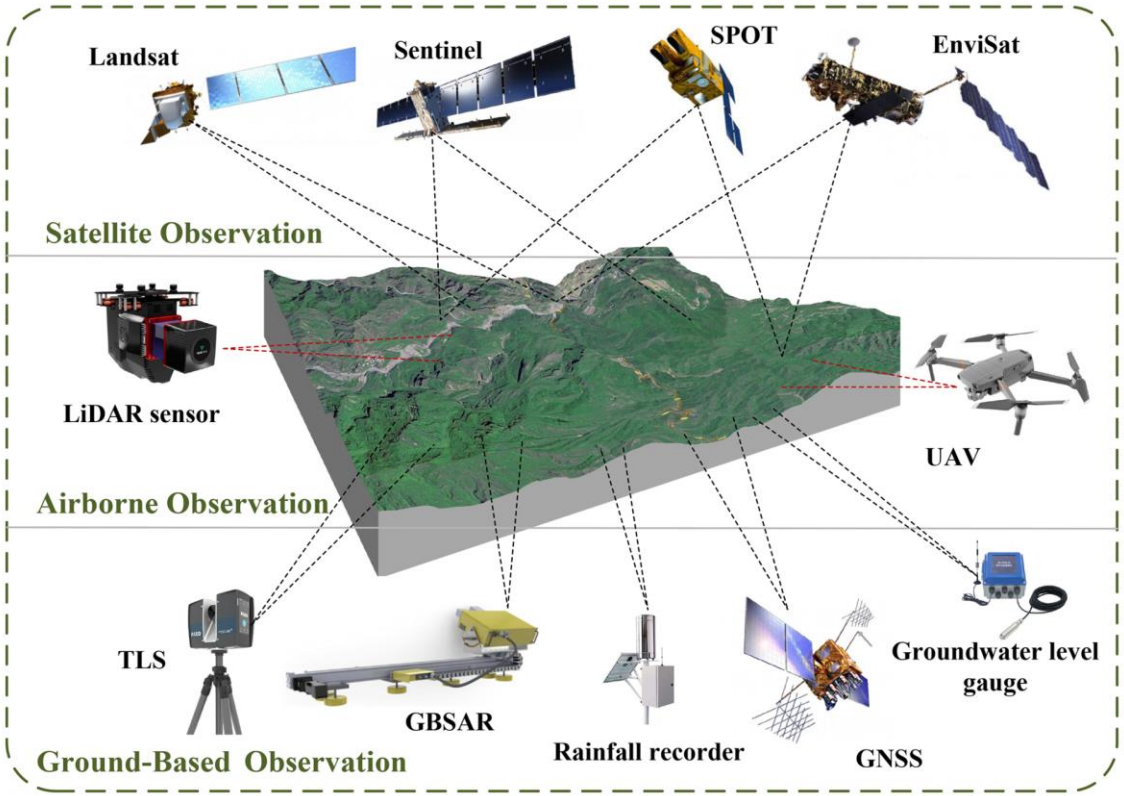
Reichstein et al. (2019)

Cartographie du sous-sol



https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2021RG000742

Détection des glissements de terrain



https://nhess.copernicus.org/articles/26/487/2026/



#02

MACHINE LEARNING ET SIMULATION NUMERIQUE

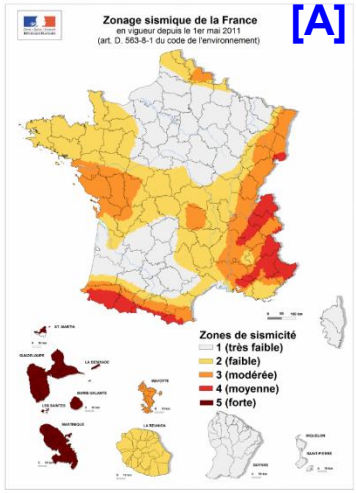
Géosciences pour une Terre durable

brgm

Big data mais pas que…

Evaluation :

- Sur une grande région (e.g., vallée, département); [A]
- En quasi ‘temps réel’ en utilisant les données accessibles [B]
- Pour des lieux difficilement accessibles (e.g., haut d’une falaise, mine profonde); [C]
- Sous contraintes financières / temps / espace …



[A]


Esterhuizen et al. 2011 [C]

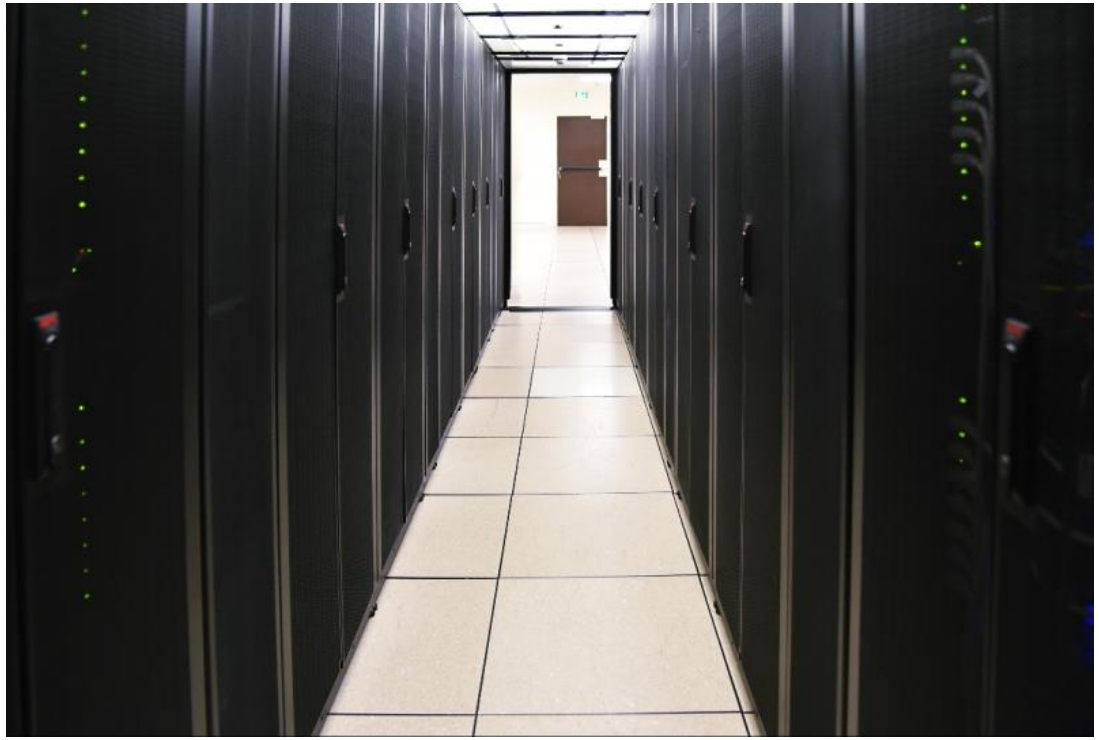

[B]

[Crédit] ARCHIVES CO



Adapté de https://www.fix-dessinateur.com/trou-dans-la-raquette/

« small data » … à cause des temps de calcul

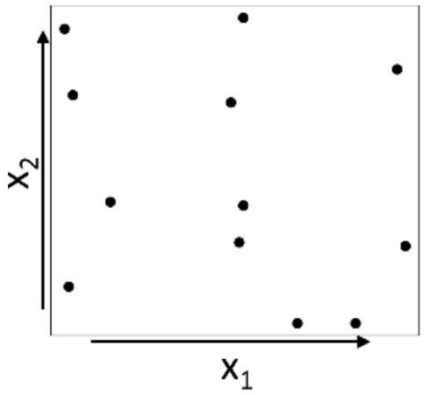


SAPHIR, le supercalculateur du BRGM, est hébergé dans son datacenter.
© BRGM - C. Boucley

	Nœud de calcul thin	Nœud de calcul Fat
Nom	hpc-thin01 à hpc-thin16	hpc-fat01 à hpc-fat06
Nombre de nœuds	16	6
Nombre de cœurs par nœud	128	128
Type de processeur	AMD Genoa 9534 2.45GHz	AMD Genoa 9534 2.45GHz
Carte Graphique	-	
Mémoire disponible par nœud (GB)	384	768
Disque local (GB)	843	843
Infiniband	HDR100	HDR100

Idée: exploiter les résultats de simulations numériques d'un nombre limité de tempêtes = base d'apprentissage

Design of Experiments (DoE)
Points to perform simulations



X1, X2 : paramètres d'entrée du simulateur numérique

Le nombre de paramètres peut être supérieur à 2 !

EXEMPLE:

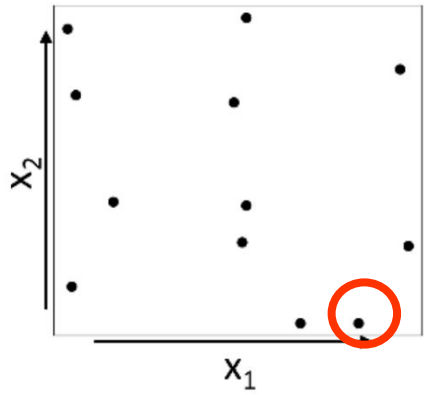
- (E, c, φ) x nombre de formations, , X,Y(source)
- Hs, Tp, Dp, U, Du, S, T, etc.
- (Vp, Vs) x nombre de formations, X,Y,Z(source)
- Densité, viscosité, conductvité thermique, capacité thermique

Couvrir l'ensemble des possibles avec un nombre limité de scénarios 100-200

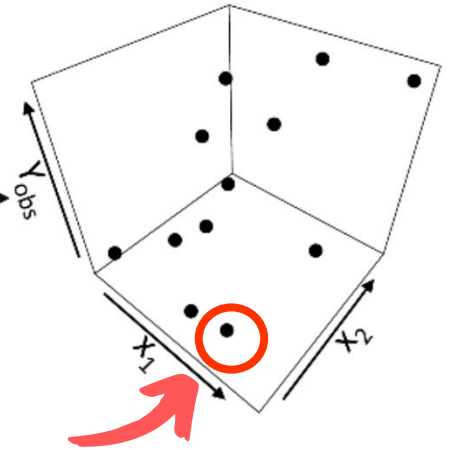
Cout de temps de calcul!

Design of Experiments (DoE)
Points to perform simulations

Simulation
Performing simulations for the points of the DoE



Computer code



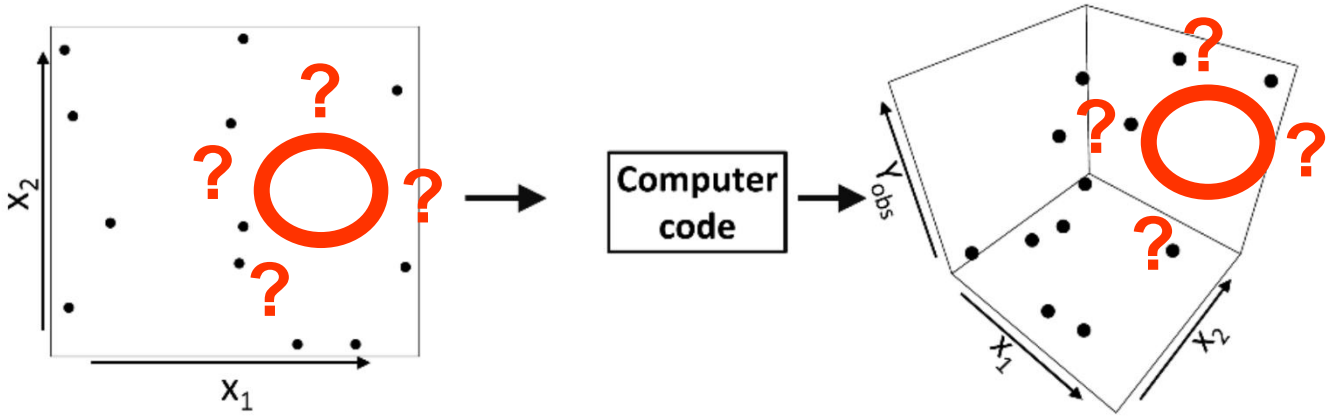
Chaque point = un résultat de simulation couteuse pour X1,X2 donné

EXEMPLE
- Distance atteinte
- Aire inondée
- PGA, PGV
- Saturation en polluant

Cout de temps de calcul!

Design of Experiments (DoE)
Points to perform simulations

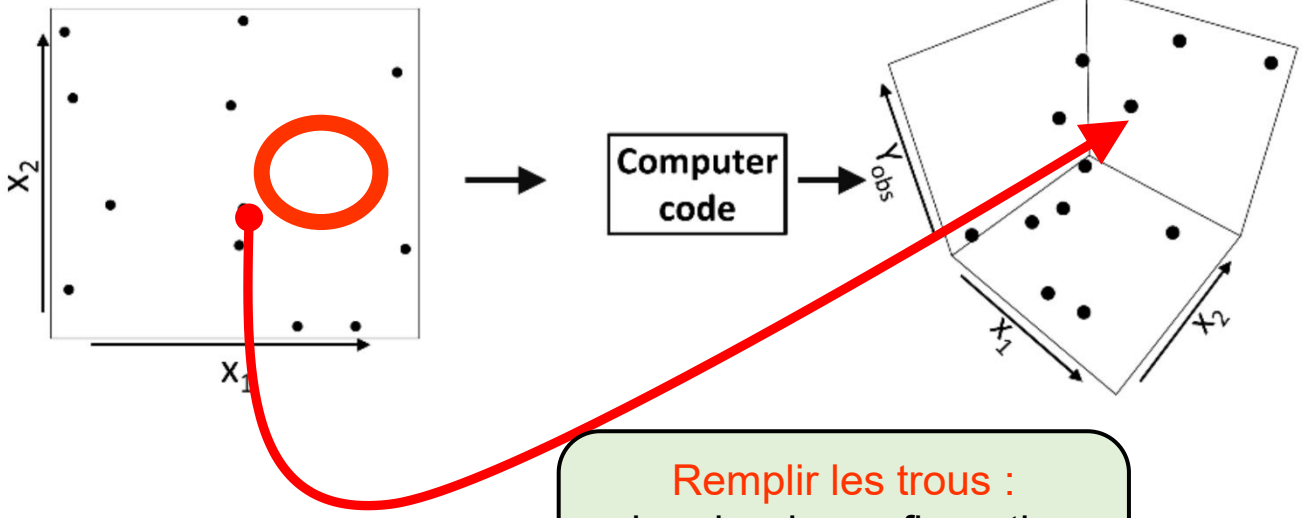
Simulation
Performing simulations for the points of the DoE





Design of Experiments (DoE)
Points to perform simulations

Simulation
Performing simulations for the points of the DoE



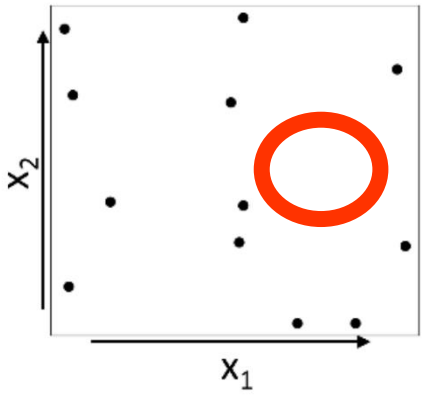
Remplir les trous :
chercher la configuration la plus proche = méthode des analogues



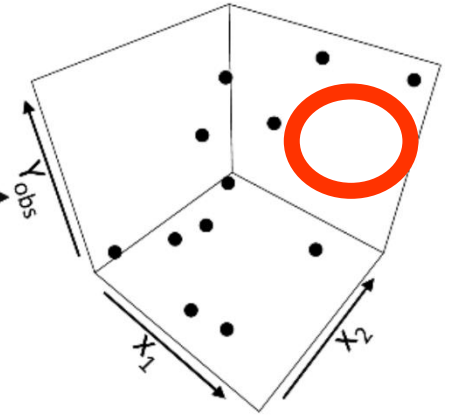
Design of Experiments (DoE)
Points to perform simulations

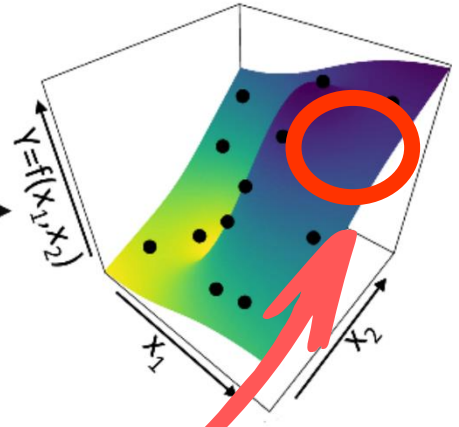
Simulation
Performing simulations for the points of the DoE

Metamodeling
Approximation of the computer code over the Xi space



Computer code





Remplir les 'trous' avec un modèle de machine learning (=métamodèle) au coût calculatoire faible

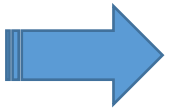
Approximation à valider !

EN PRATIQUE

Métamodèle = modèle de substitution = proxy = surface de réponse = modèle **statistique** construit à partir d'un **nombre limité** de résultats de simulations numériques

Ingrédients

- Une base de résultats de simulations numériques.
  - Etre capable de **lancer >100 simulations** en (semi-)automatique
  - **Stocker / traiter** les résultats en (semi-)automatique
  - Etre capable **d'enrichir** la base de résultats



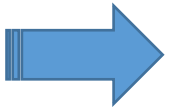
Un métamodèle est lié à un modèle numérique construit pour une étude / un site

EN PRATIQUE Métamodèle = modèle de substitution = proxy = surface de réponse = modèle **statistique** construit à partir d'un **nombre limité** de résultats de simulations numériques

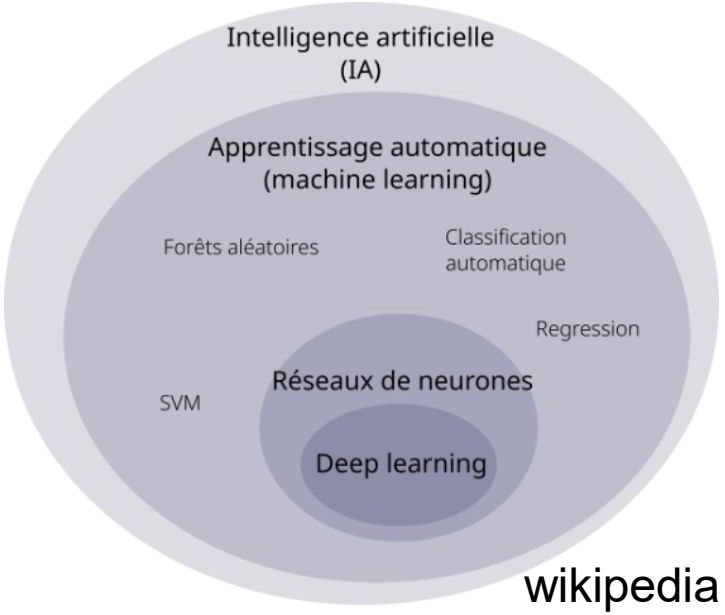
Ingrédients

- Une base de résultats de simulations numériques.
  - Etre capable de **lancer >100 simulations** en (semi-)automatique
  - **Stocker / traiter** les résultats en (semi-)automatique
  - Etre capable **d'enrichir** la base de résultats

- **Modéliser la relation entrées – sortie**
  - Outils **statistiques** et **d'apprentissage machine:**
    - modèle de régression, modèle d'interpolation, arbres de décision, réseau de neurones, deep learning, etc.
  - **Valider la substitution**

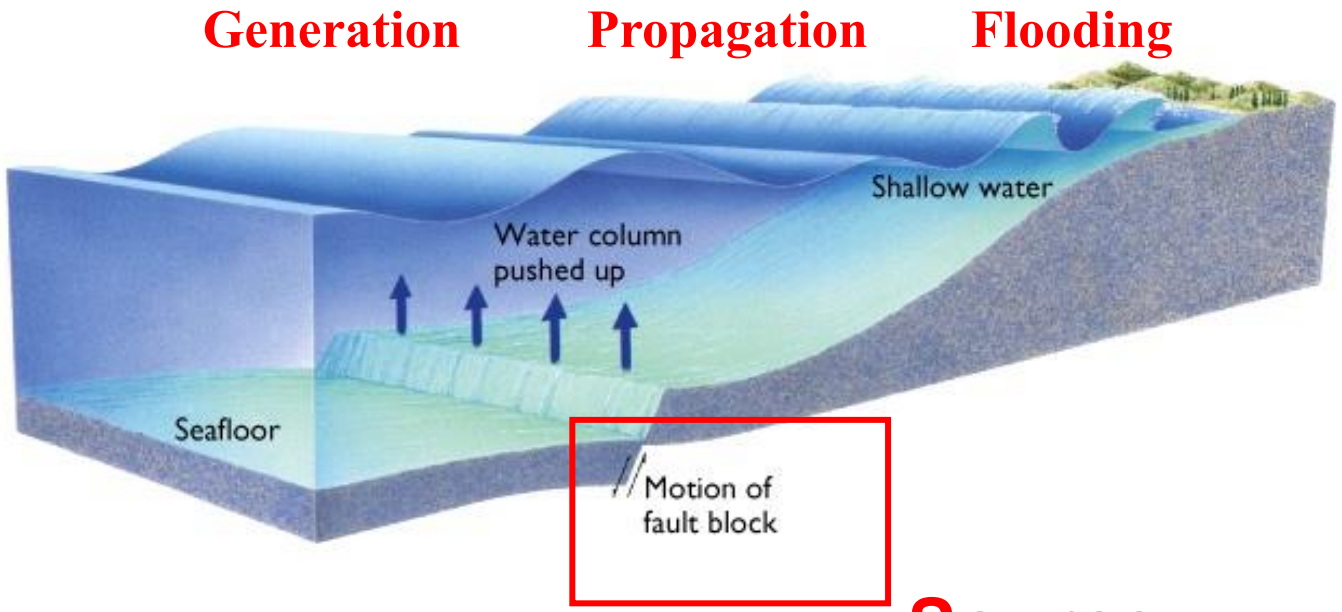


Capacité prédictive + Incertitude



wikipedia

Example of Earthquake-induced tsunami



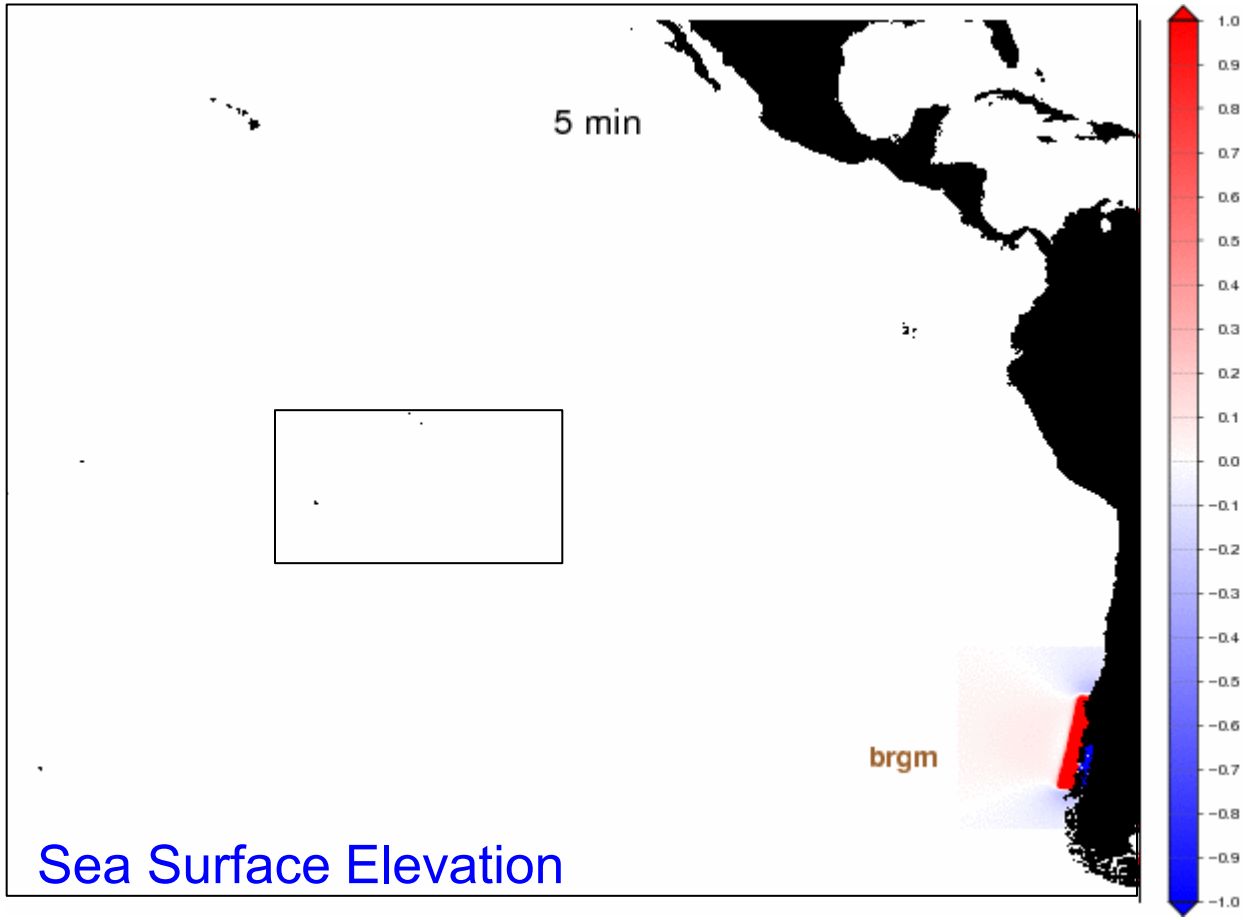


Source

Location, geometry, fault displacement, etc.

Simulation of the tsunami generated in Chili in 1960

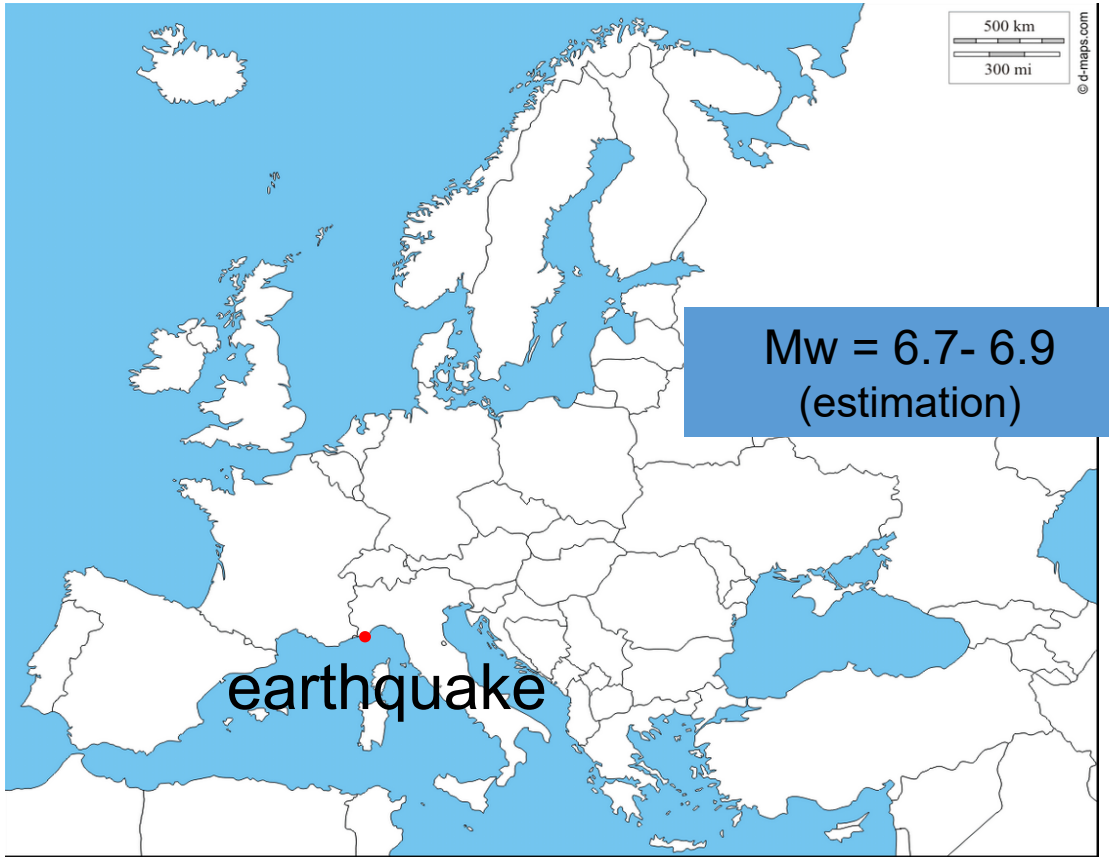
Generation (Okada 1985) and propagation (Funwave-TVD) up to French Polynesia - **R. Pedreros (BRGM)**



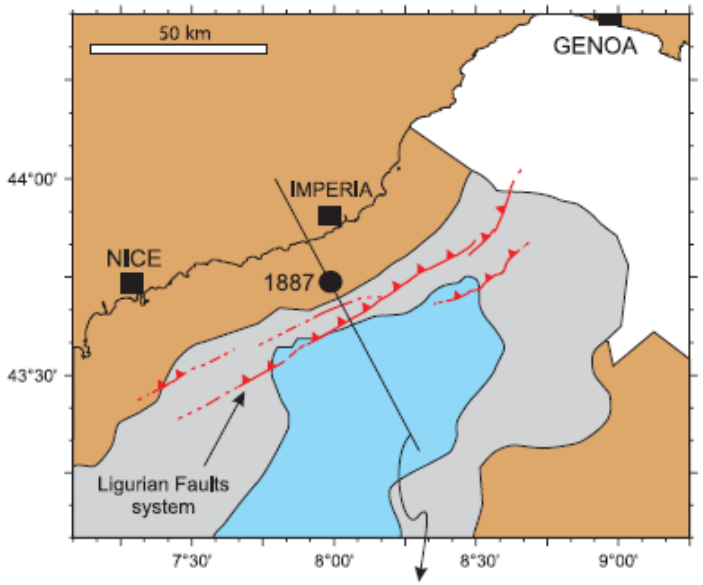
Sea Surface Elevation

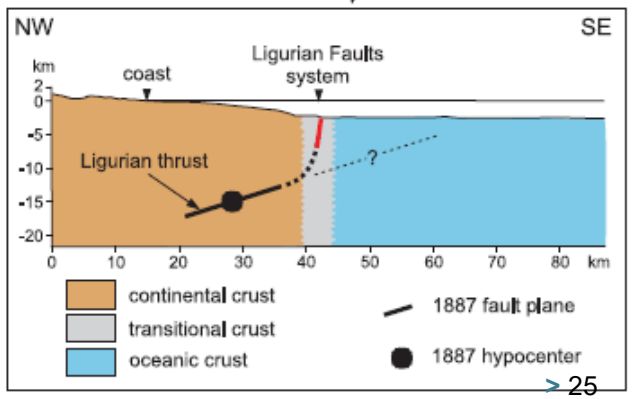
Motivating case study: The 1887 Ligurian event

Damaging historical earthquake occuring at the junction between the southern French-Italian Alps and Ligurian basin
induced tsunami



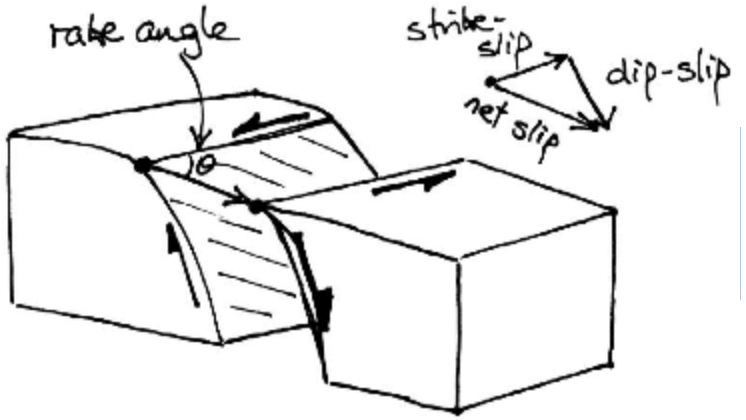
Mw = 6.7- 6.9
(estimation)





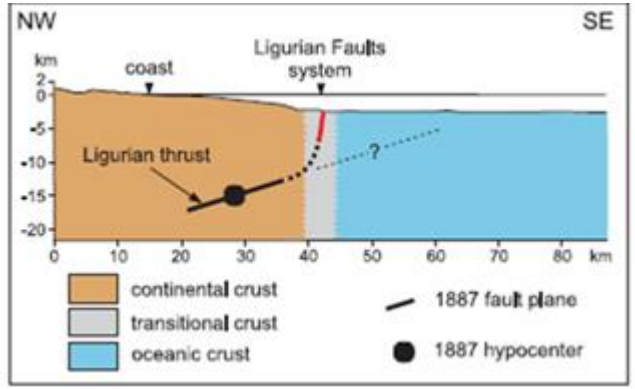
Larroque et al. 2012

Uncertainty characterisation of the source



Location			orientation				geometry	
Lon (°)	Lat (°)	Depth Z (km)	Azim (°) 2 scenarios North **(South)** dipping	DIP (°)	Rake (°)	Average displacement D (m)	Length (m)	Width (km)
7.78	43.45	5	220 **(40)**	10	70	0,3	20	10
8.15	43.92	20	260 **(80)**	70	110	2	57	22

Based on Larroque et al. (2012) and Ioualalen et al. (2014)

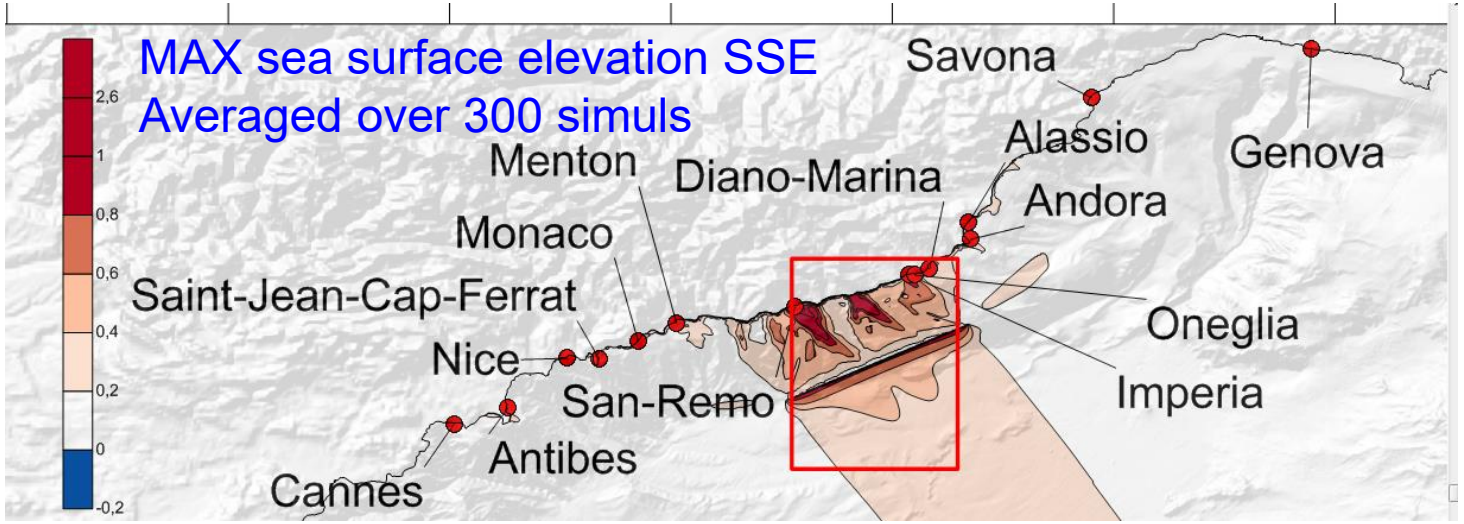


Numerical simulation

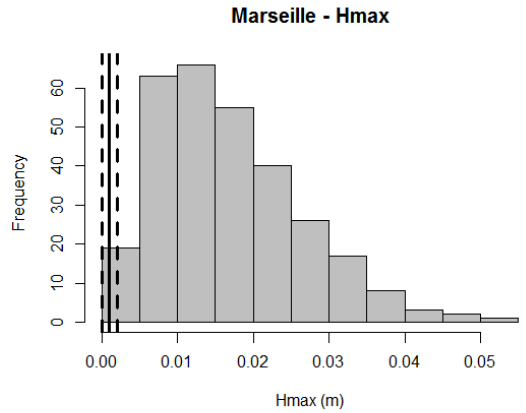
MODEL CHARACTERISTICS

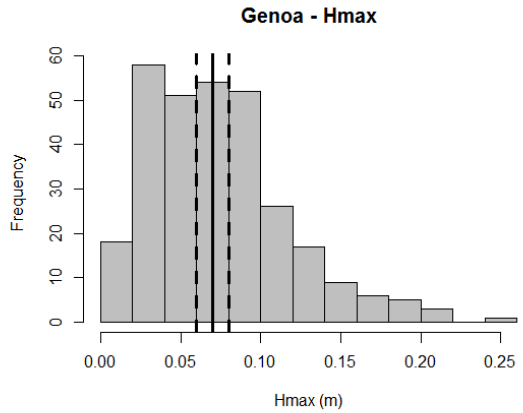
- 2,650,000 mesh cells
- CPU time of a single simulation ***30-60 min*** with ***4*24 cores***
- Funwave_TVD Boussinesq ocean surface wave propagation + Okada for initial deformation

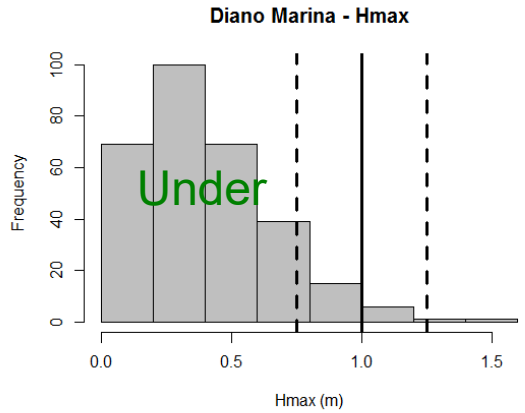
300 configurations were randomly generated (latin hypercube sampling and uniform distribution).

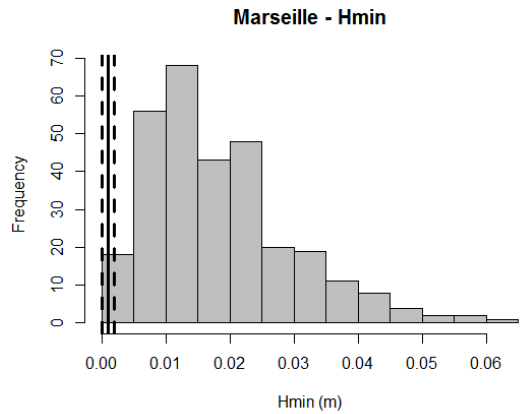


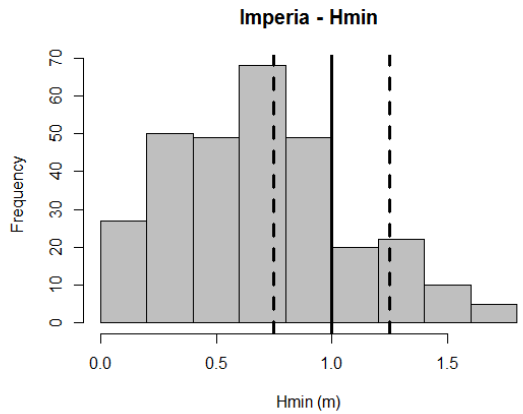
Simulation results (a priori)

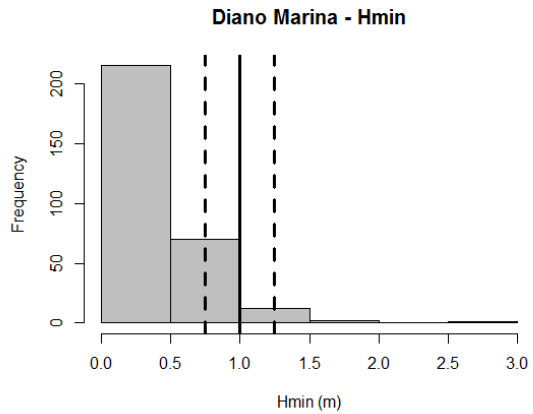










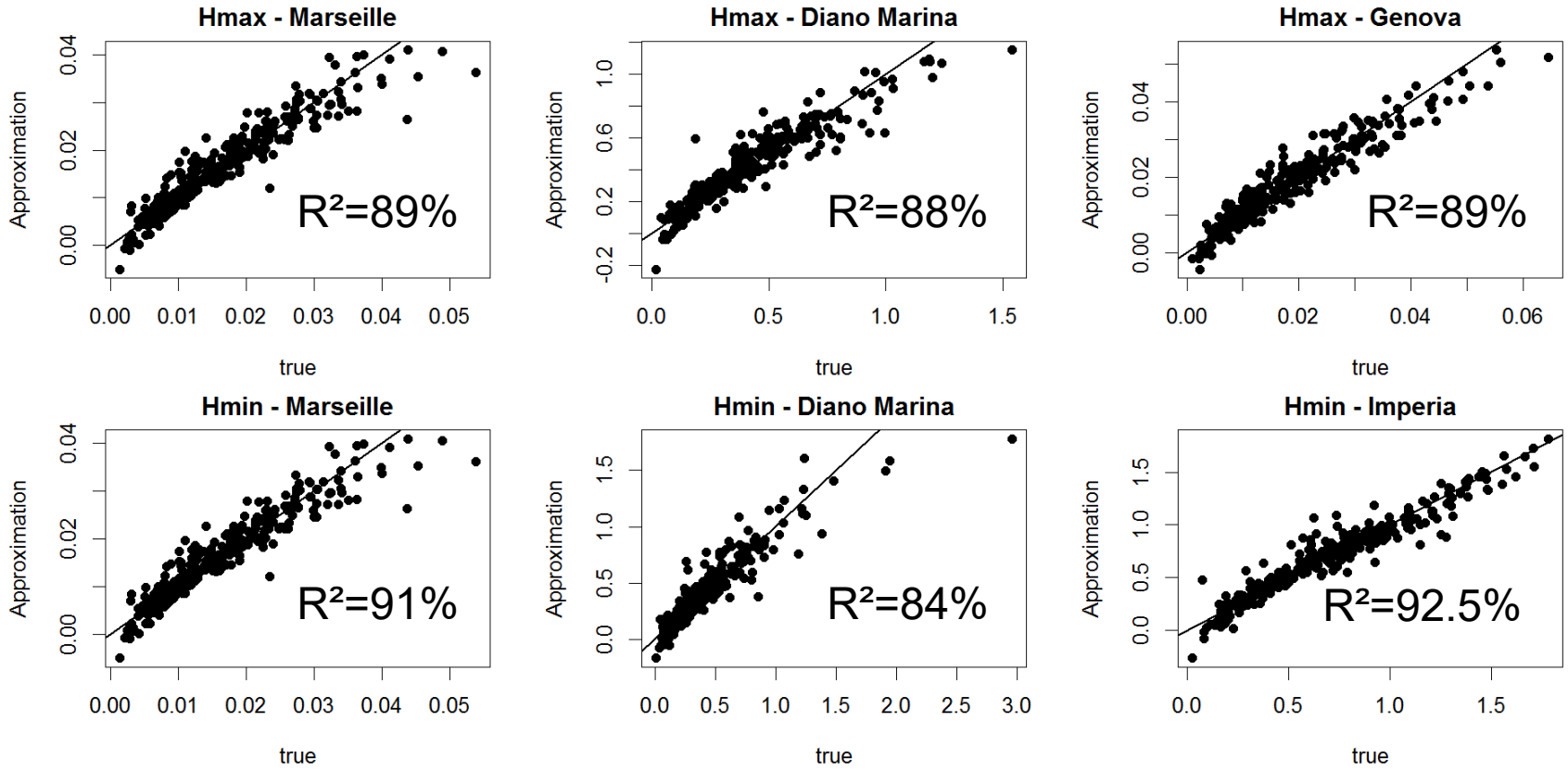


Observation

Assumed range of uncertainty

Overcoming the computation time cost with kriging (Gaussian process) surrogate models

training with 300 simulation results

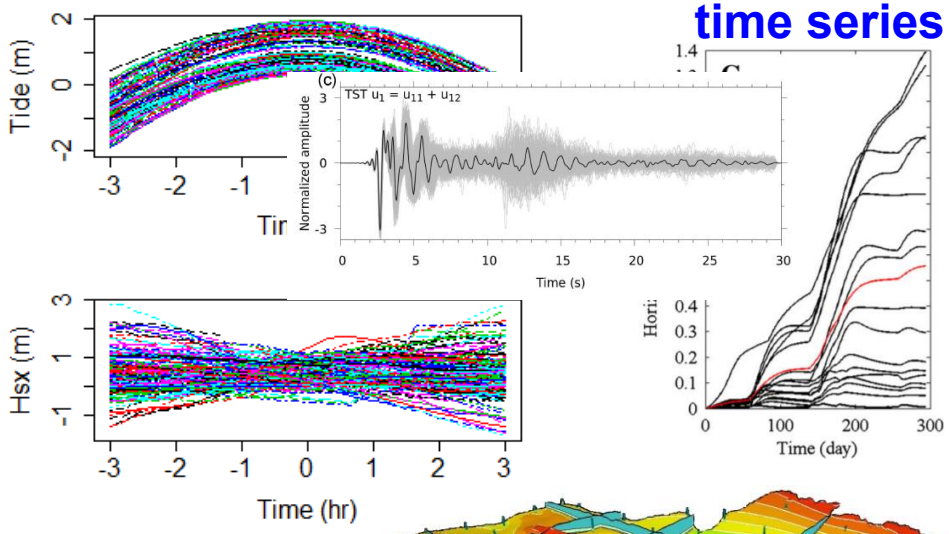


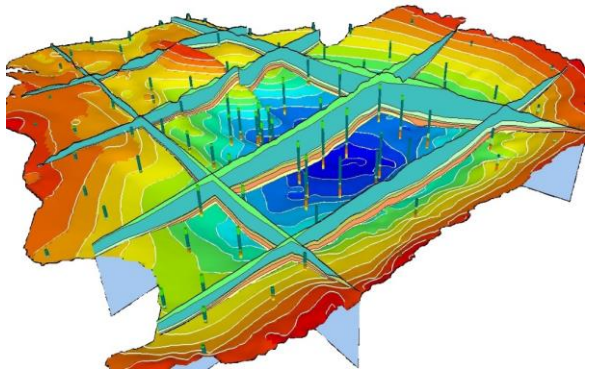
Satisfactory approximation quality (typical threshold at 80%, see e.g., Marrel et al., (2009); Storlie et al. (2009)



Faire des statistiques avec des objets haute résolution (séries temporelles, cartes, etc.) quand le nombre d'observations est limité ?

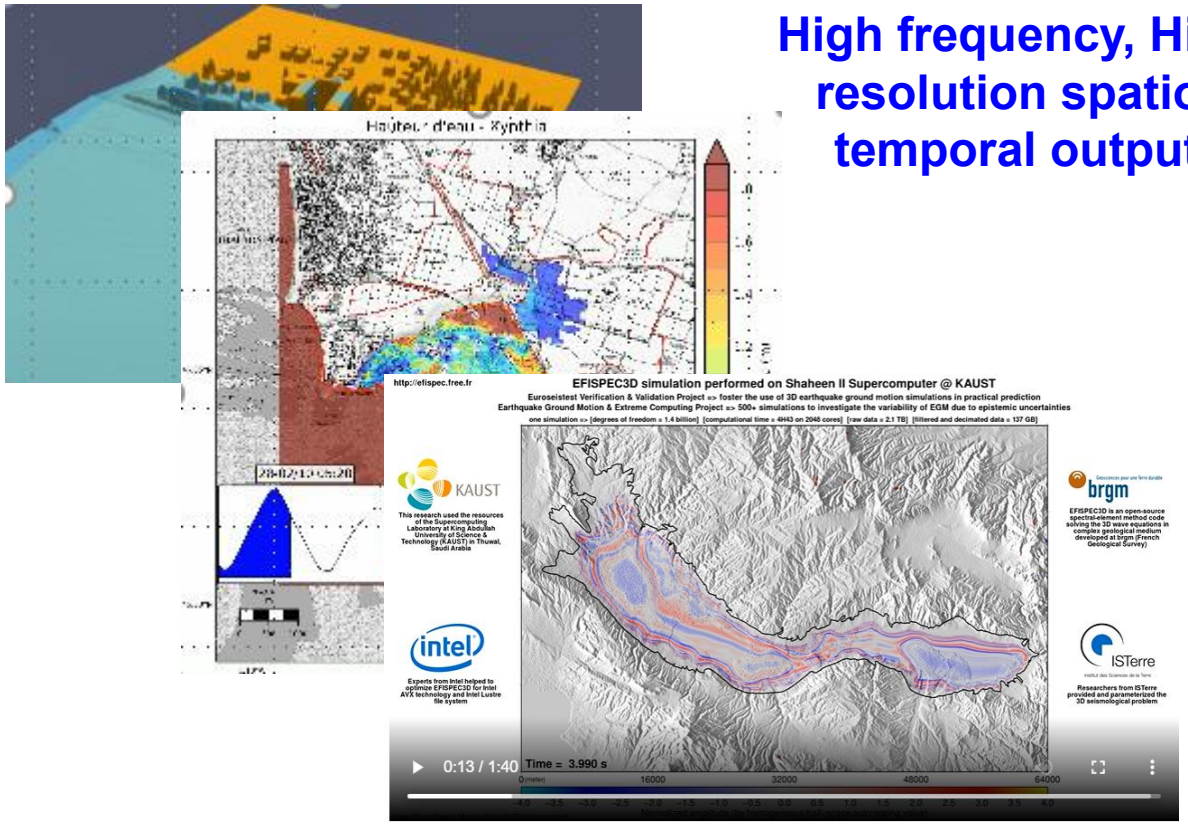
High frequency input time series



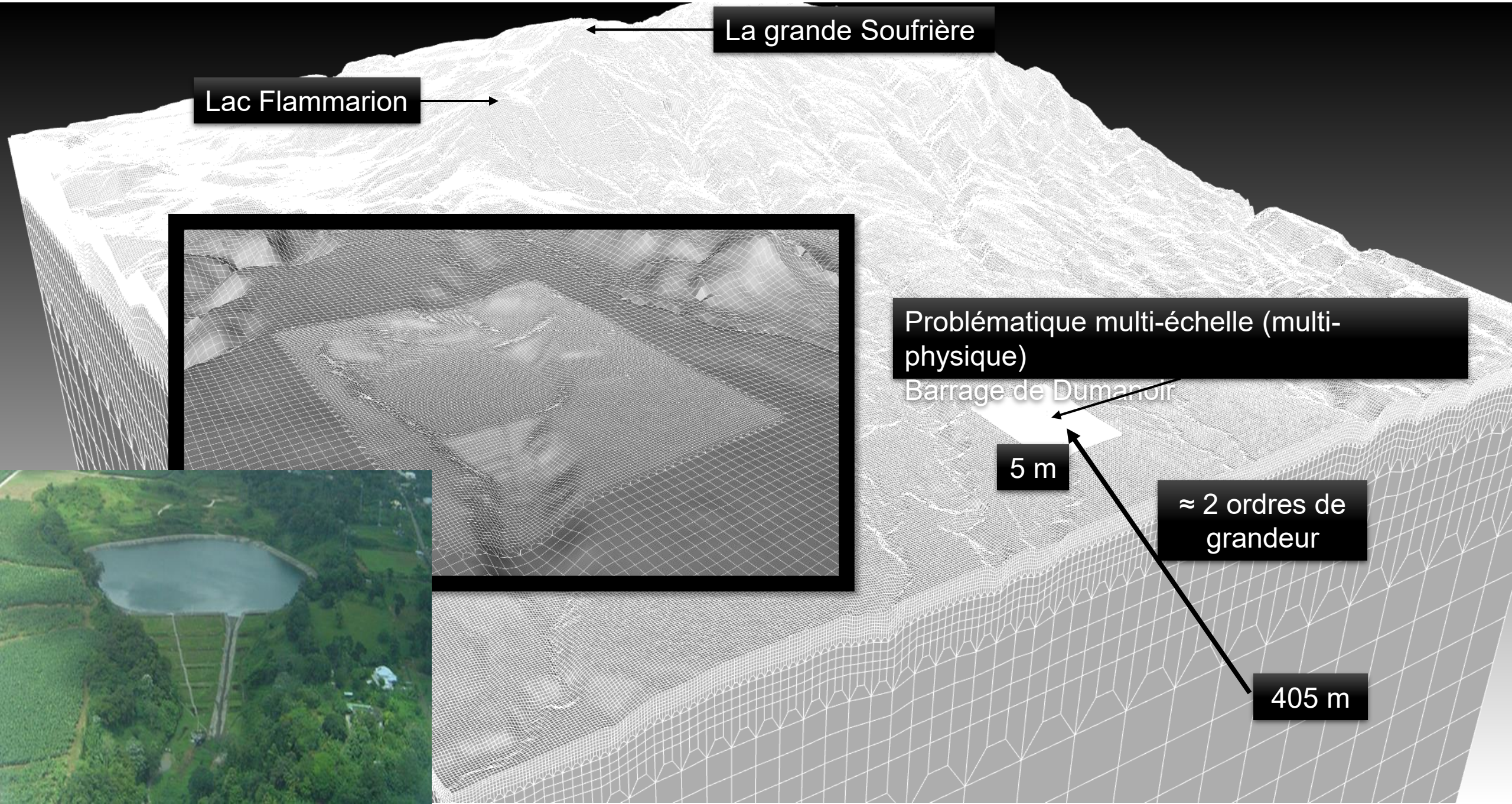


High resolution input spatial field

High frequency, High resolution spatio-temporal output

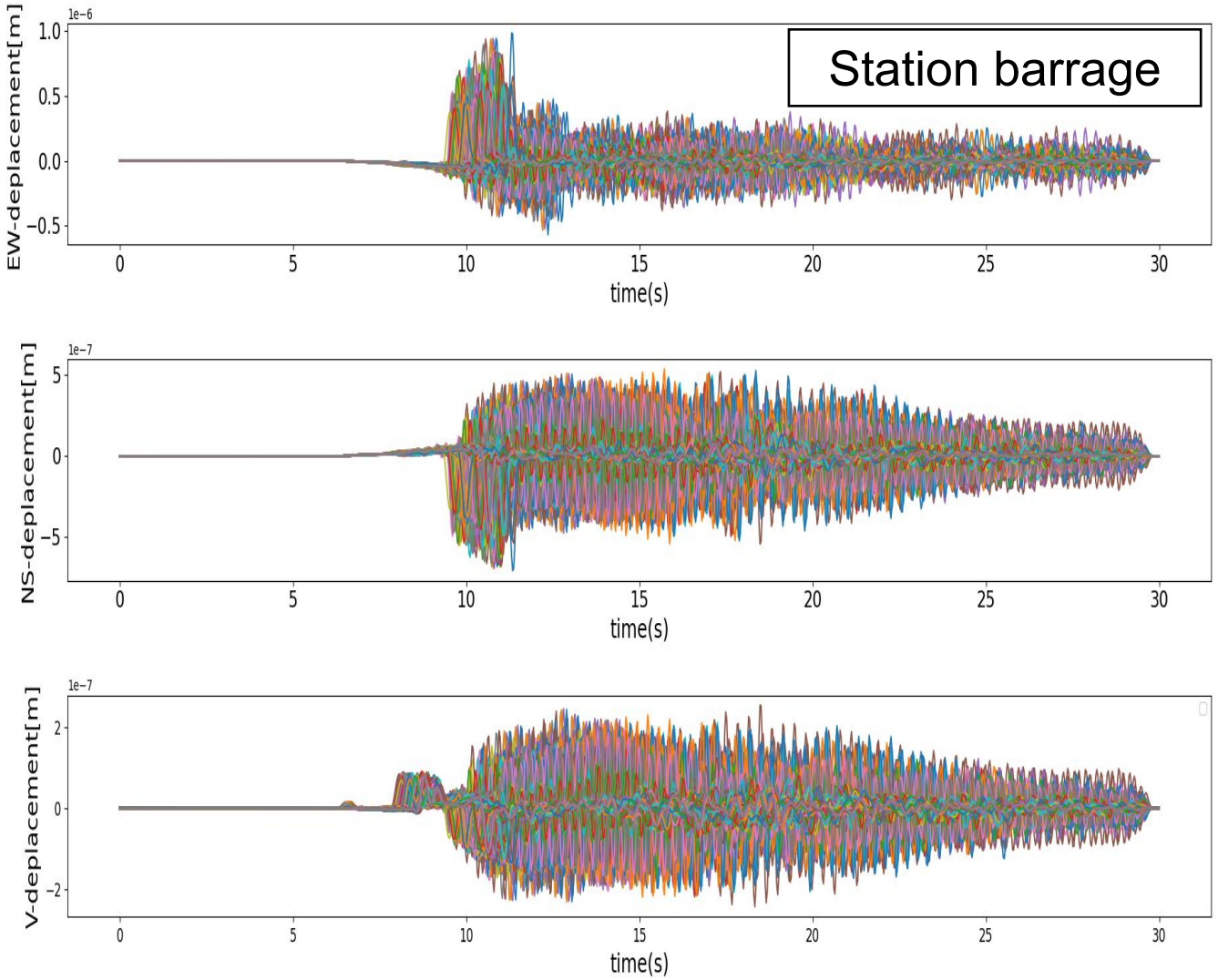


Thèse de Mohamad Ali Noureddine – prédire en temps



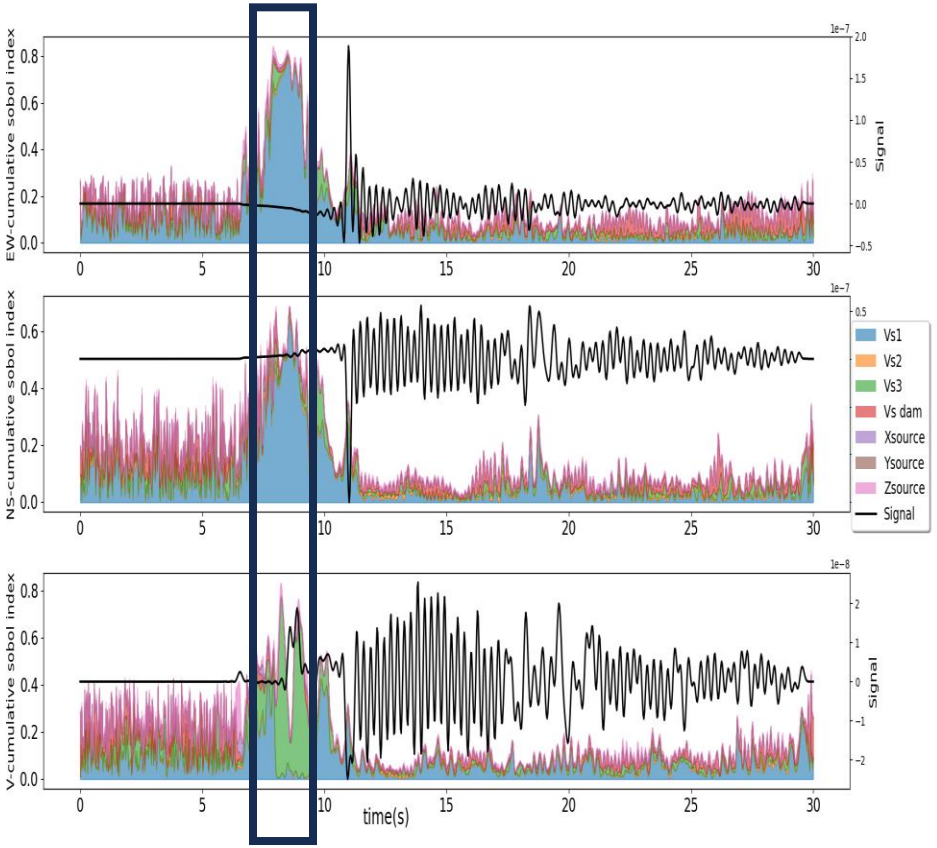
Thèse de Mohamad Ali Noureddine – prédire en temps

600 simulations : 200 000 heures monocoeur



NN simulations : millisecondes

Analyse globale de sensibilité



- *Méta modèle de sismogrammes basé sur un réseau neuronal NN pour la quantification de l'incertitude : Vers un jumeau numérique sismique*

A few references to go beyond the scalar-on-scalar case

- Adaptative design of experiments [Bect et al. 2012]
- Scenario-like, categorical inputs [Roustant et al., 2020; Rohmer et al. 2022]
- Complex inputs: time series [Betancourt et al., 2020], spatial inputs [Liu and Guillas 2016; Rohmer 2015]
- Complex output [El Garroussi et al., 2022; Chang et al., 2014; Perrin et al. 2020; Li et al., 2020; Gu and Berger, 2016]
- Mixing wtih low fildelity simulations [Rohmer et al., 2023; Thenon et al. 2016]
- For calibration [Higdon et al., 2008; Salter et al. 2019]



#03

QUELQUES MODÈLES POPULAIRES

Géosciences pour une Terre durable
brgm



the data

A set of samples $\mathbf{x}_i$ and labels $\mathbf{y}_i$ to learn from:

$$\mathcal{D} = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^{N}$$



the model

A parametric function f_θ that maps data $\mathbf{x}$ to $\hat{\mathbf{y}}$

$$f_\theta : \mathbf{x} \mapsto \hat{\mathbf{y}}$$



the loss

A measurement of the model performance

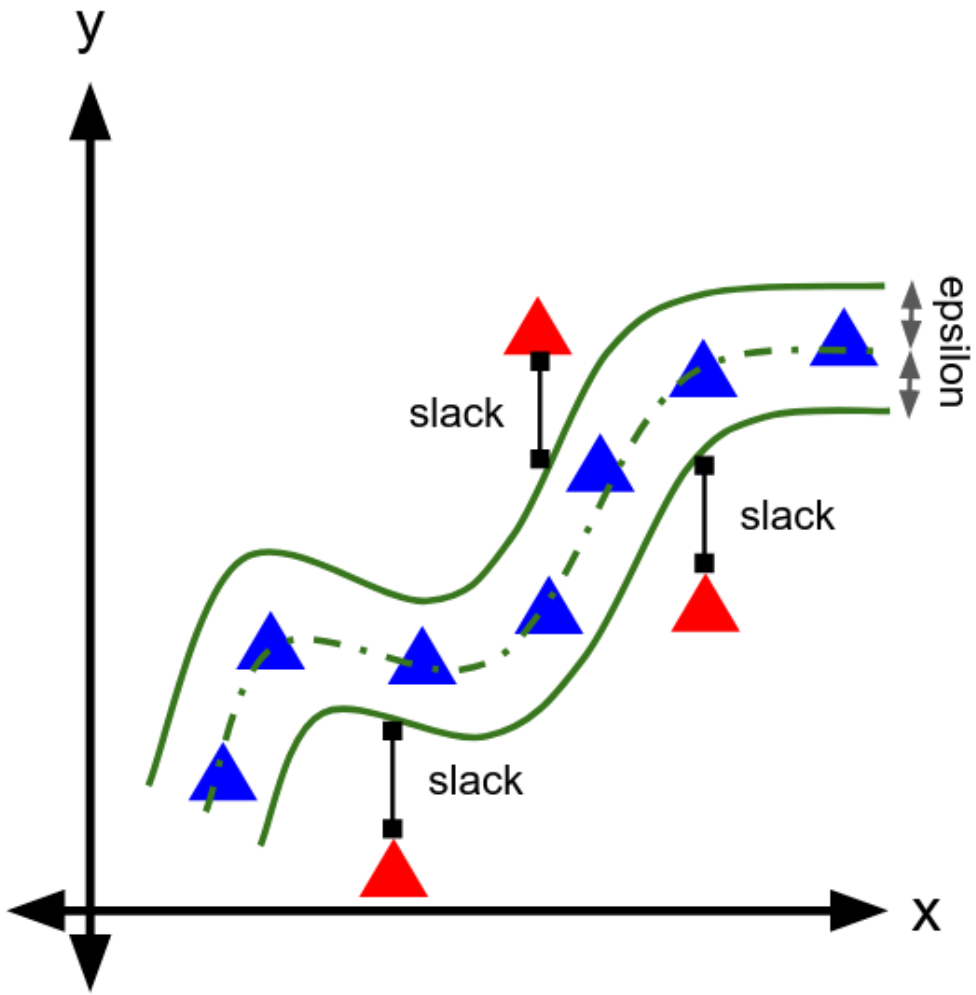
$$\mathcal{L}(\hat{\mathbf{y}}, \mathbf{y})$$

Learning = find the optimal parameters θ^* that minimize the loss $\mathcal{L}$ function

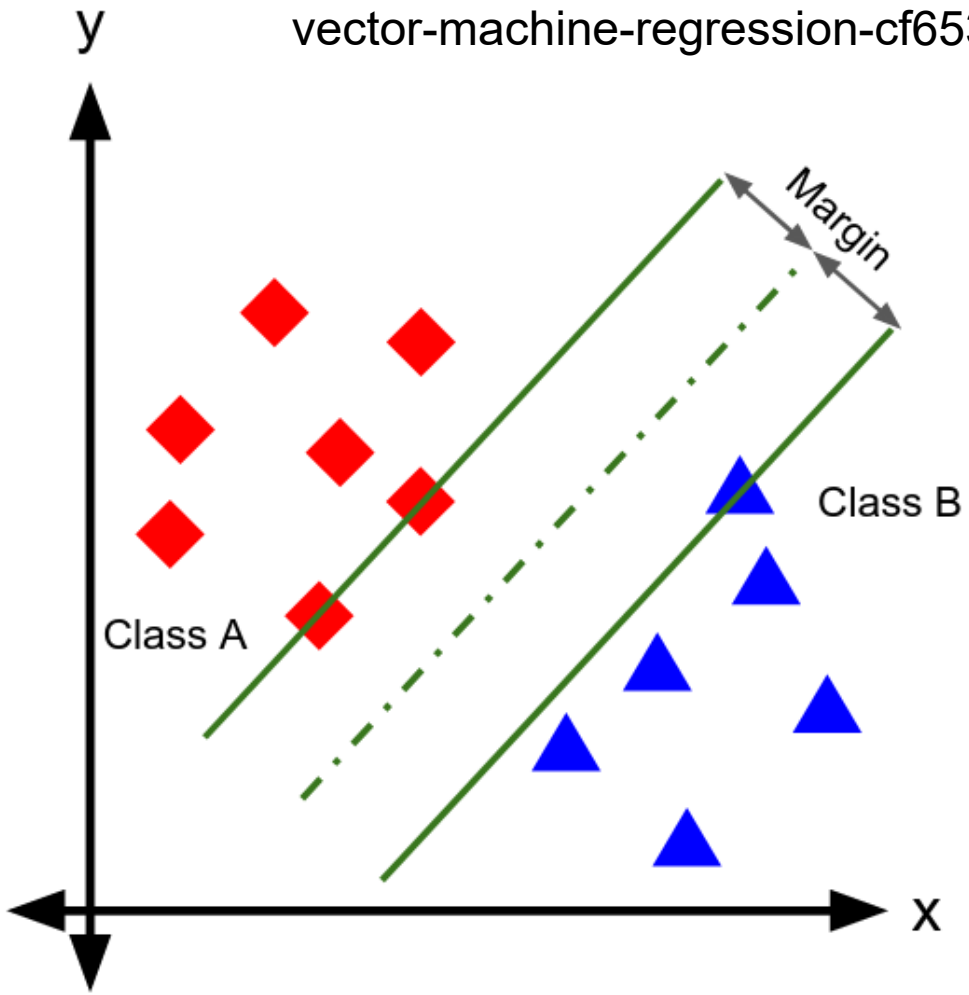
$$\theta^* = \underset{\theta}{\operatorname{argmin}}\, \mathcal{L}\Big(f_\theta(\mathbf{x}), \mathbf{y}\Big)$$

Support Vector Machine for regression

https://medium.com/it-paragon/support-vector-machine-regression-cf65348b6345



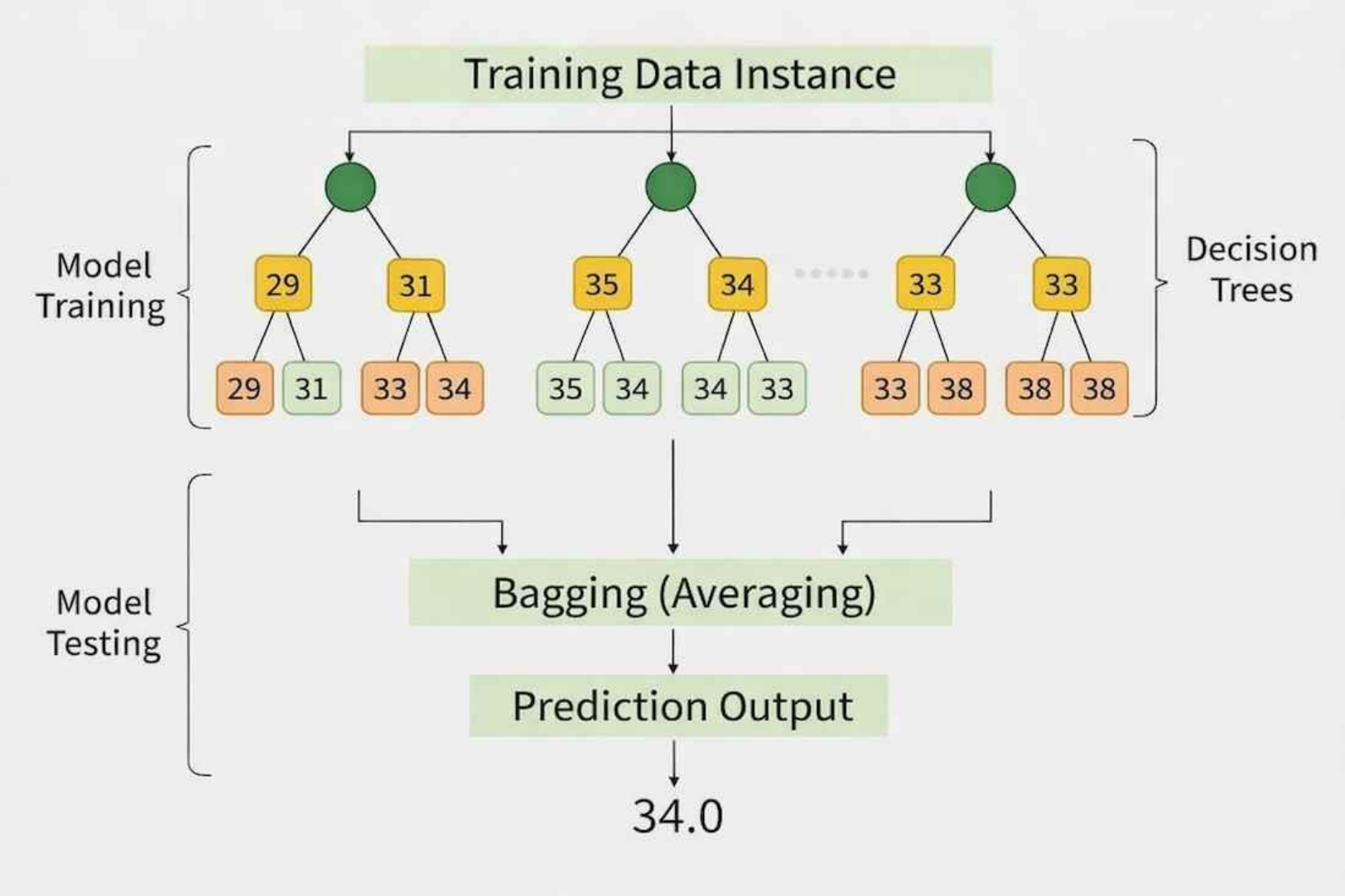
Regression



Classification

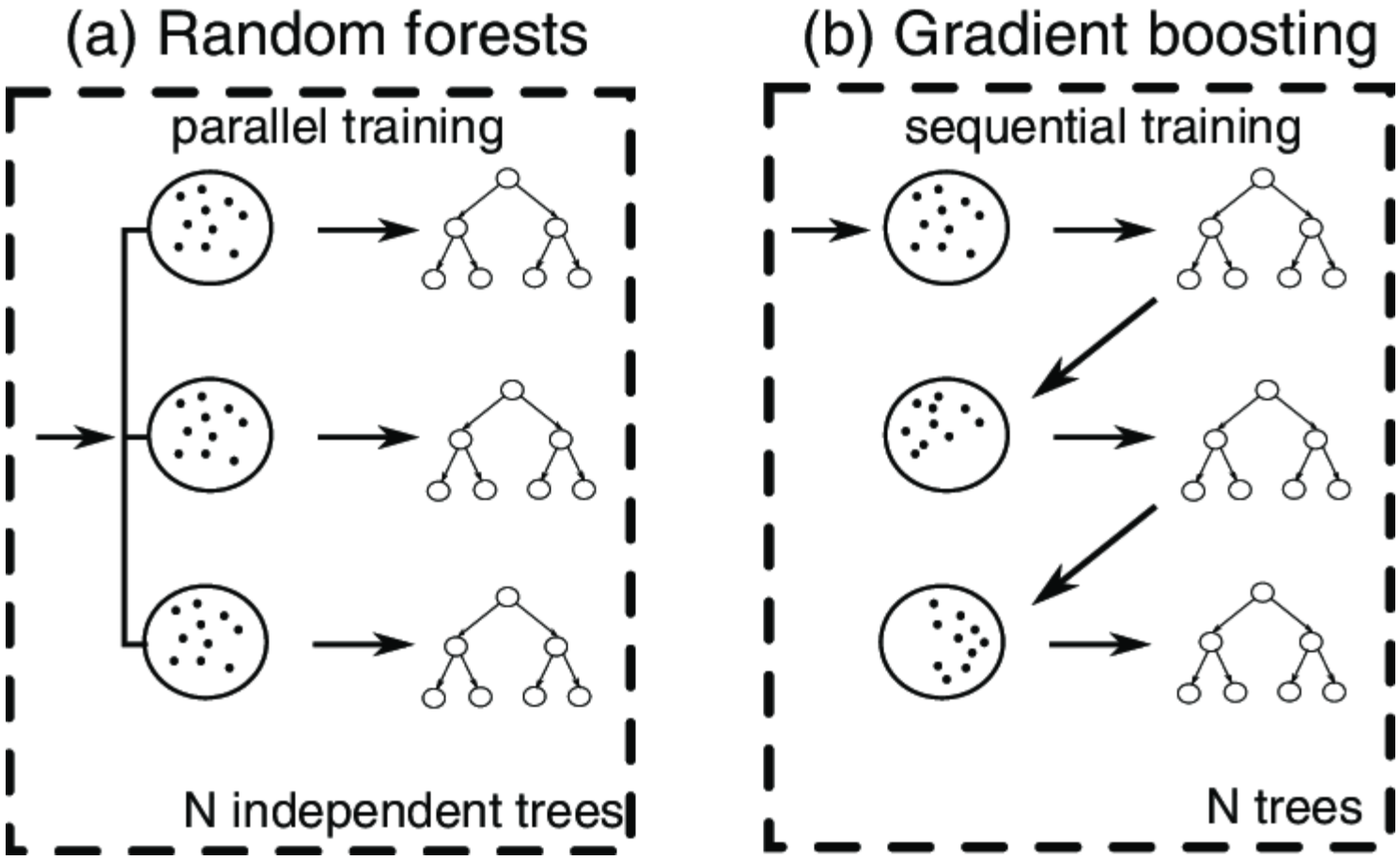
https://www.geeksforgeeks.org/machine-learning/random-forest-regression-in-python/

Random forest

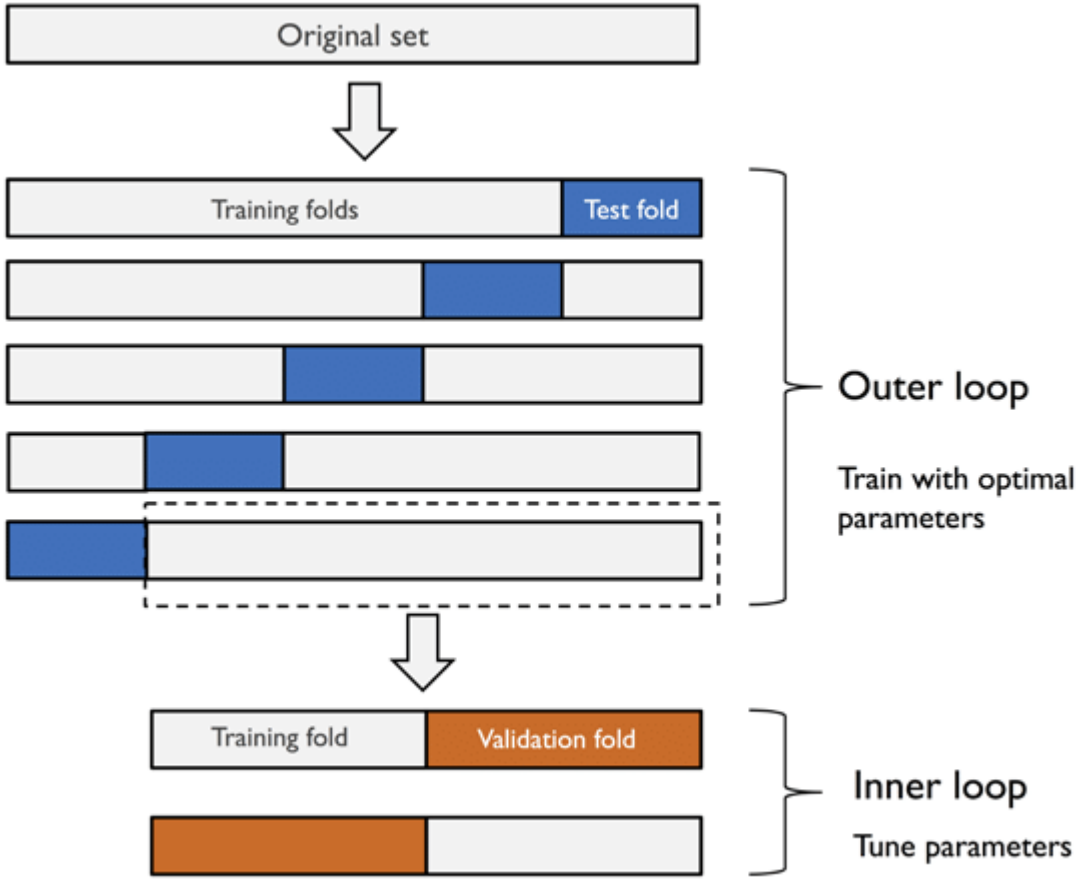


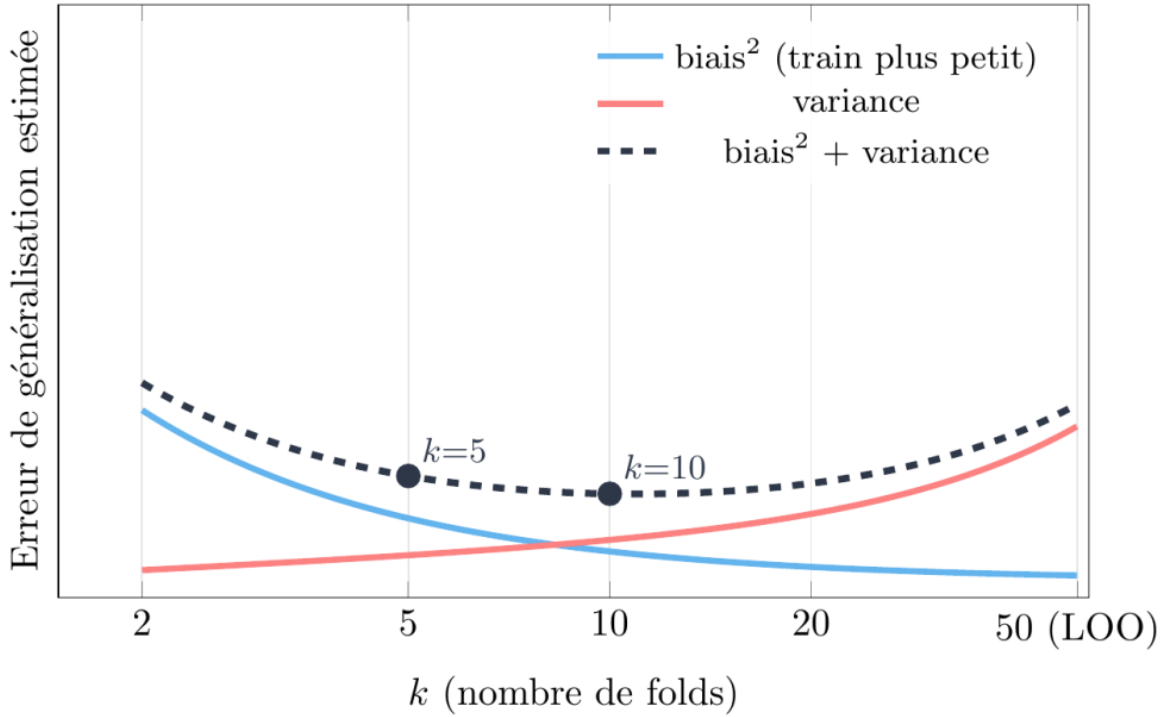
https://doi.org/10.1103/PhysRevE.100.032410

Xgboost



Validation croisée



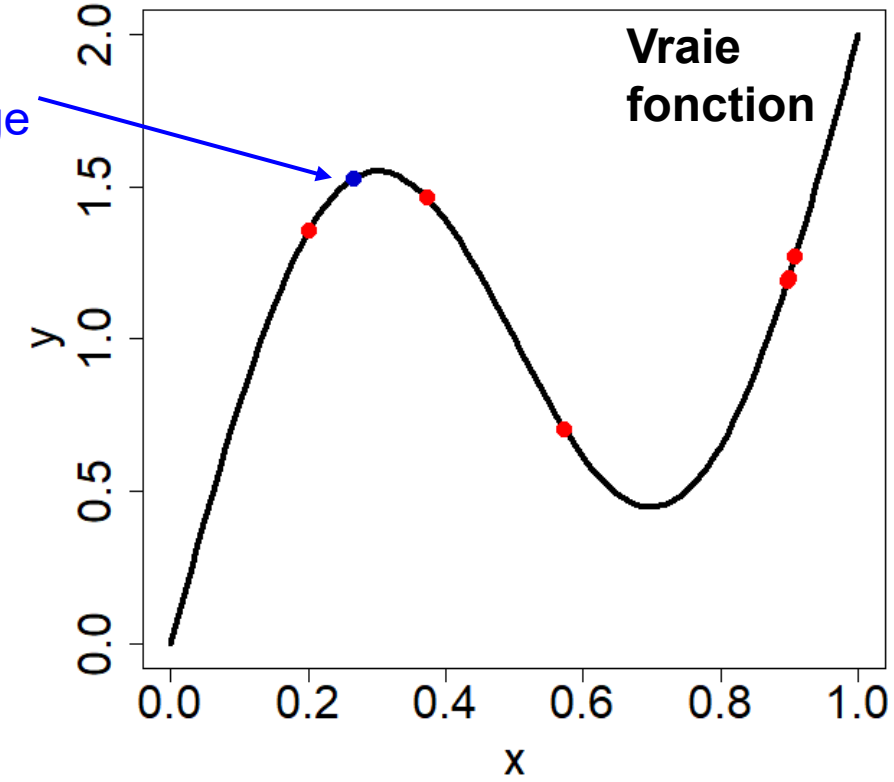


https://www.researchgate.net/figure/Exemple-de-validation-croisee-imbriquee-de-type-5x2-cross-validation-Raschka-and_fig39_331588166

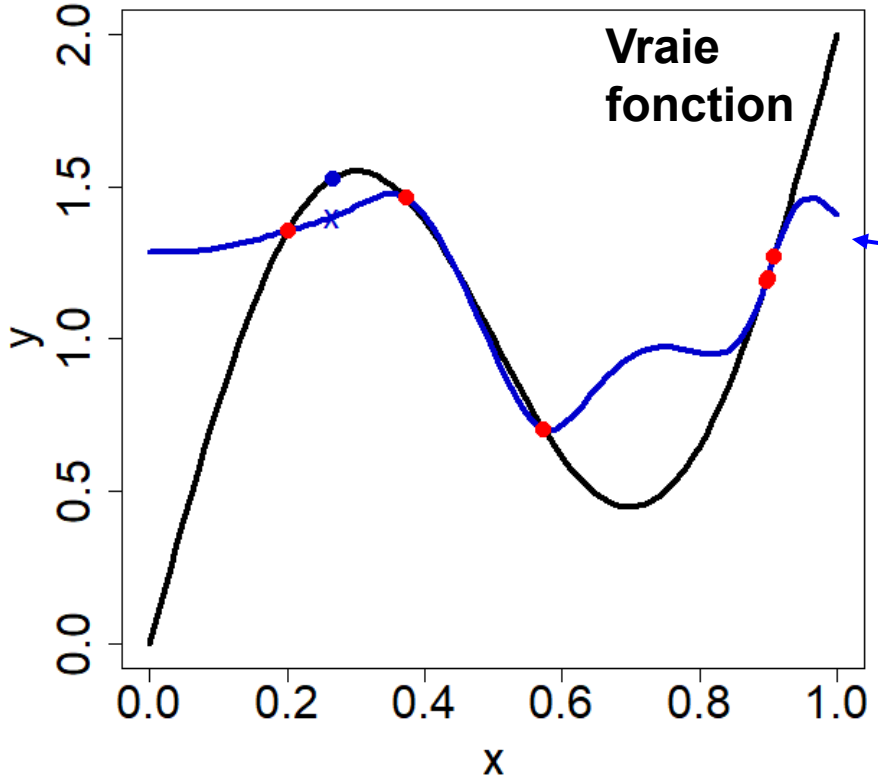
https://progresser-en-maths.com/validation-croisee-cours-et-applications-ml/

Principe de la validation croisée ’Leave-One-Out’

Extraction d’une observation de la base d’apprentissage



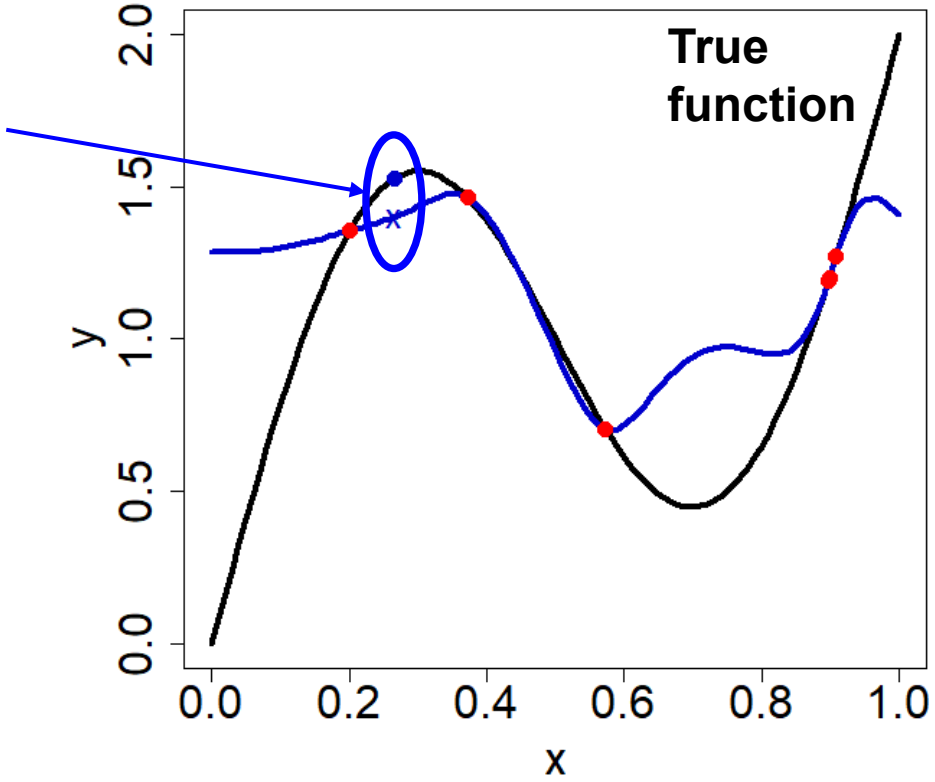
Principe de la validation croisée ’Leave-One-Out’



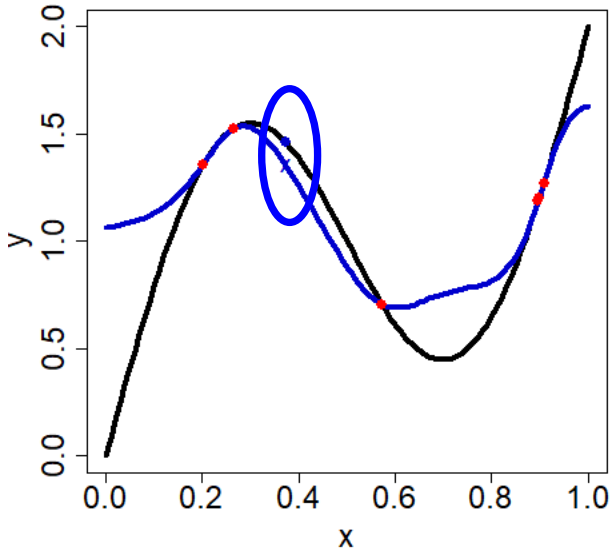
Mise à jour du modèle ML

Principe de la validation croisée ’Leave-One-Out’

Estimation de l’erreur



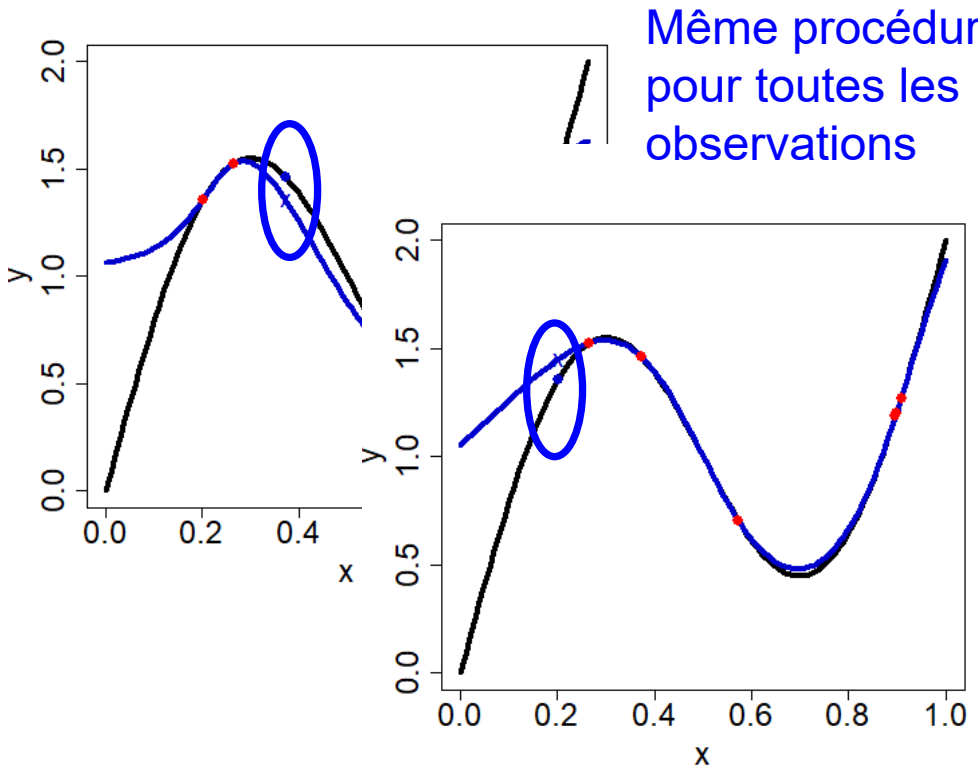
Principe de la validation croisée ’Leave-One-Out’



Même procédure pour toutes les observations

etc….

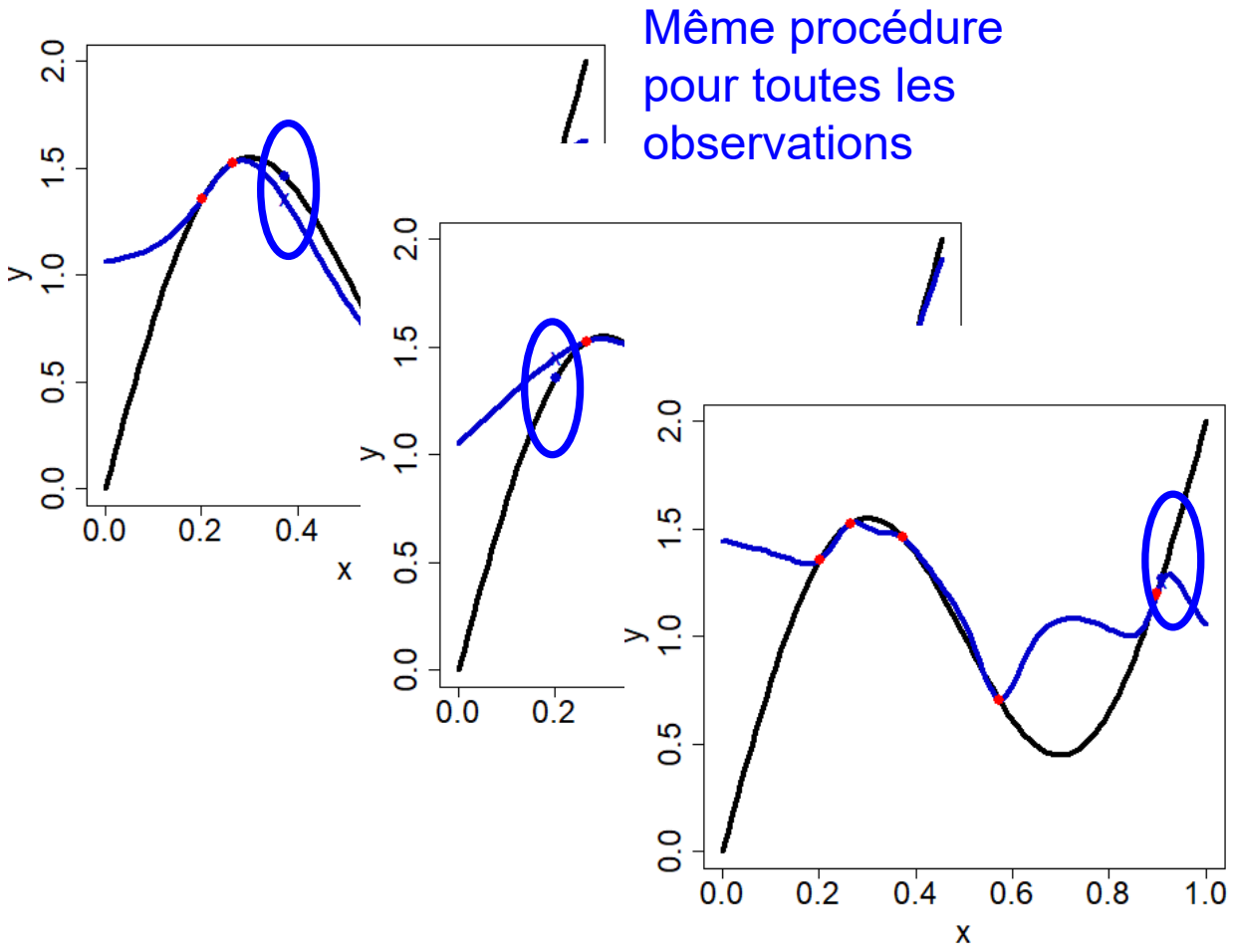
Principe de la validation croisée ’Leave-One-Out’



Même procédure pour toutes les observations

etc….

Principe de la validation croisée ’Leave-One-Out’

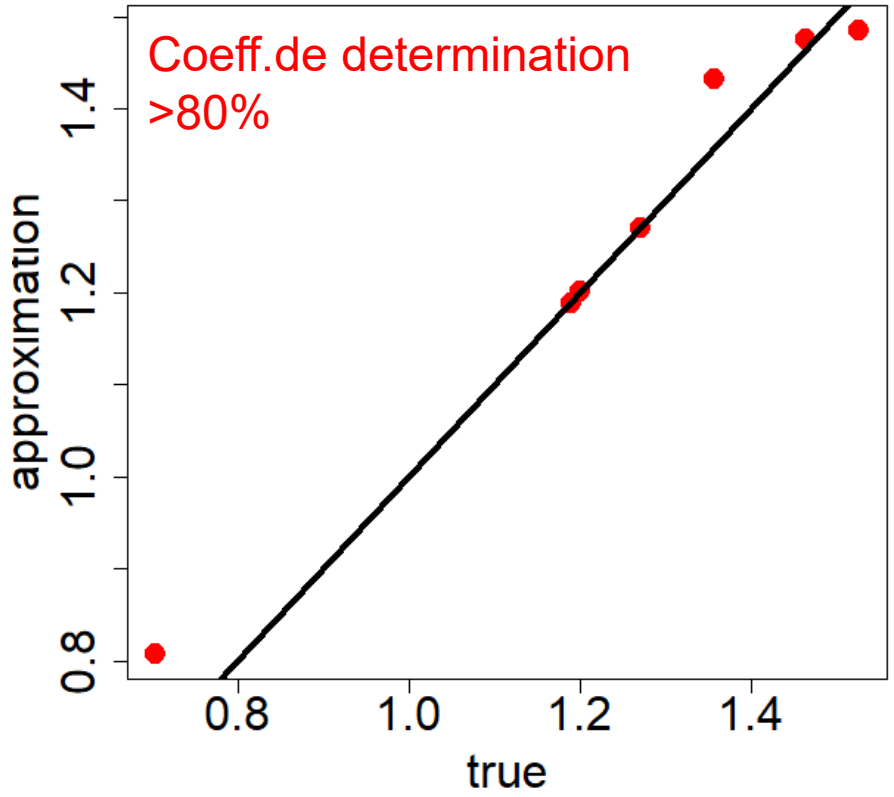


Même procédure pour toutes les observations

etc….

Principe de la validation croisée ’Leave-One-Out’

$$Q^2 = 1 - \frac{\sum_{i \in \boldsymbol{T}} (y_i - \hat{y}_i)^2}{\sum_{i \in \boldsymbol{T}} (y_i - \bar{y})^2}$$



Merci pour votre attention!
Des questions?

Contact: j.rohmer@brgm.fr